SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – I : RAILWAY ENGINEERING

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Introduction – Permanent Way Components – Cross Section of Permanent Way – Functions and Requirements of Rails, Sleepers and Ballast – Types of Gauges – Creep of Rails – Theories Related to Creep – Coning of Wheels – Adzing of Sleepers – Rail Fastenings

1. Introduction to Railway Engineering

Railway = a guided land transport system in which vehicles (rolling stock) run on steel rails. It is the cheapest mode for **bulk goods** and **long-distance mass passenger** movement.

- First train in India: **16 April 1853**, Bombay (Bori Bunder) to Thane, 34 km.
- Indian Railways is one of the largest networks in the world (about 68,000 route-km), run by the Ministry of Railways through the Railway Board and 18 zonal railways.
- A railway system has 4 parts: **(i) track / permanent way, (ii) rolling stock (locomotives, coaches, wagons), (iii) signalling & interlocking, (iv) stations & yards.**

Advantages of railways

- Very economical for heavy and bulk transport; low rolling resistance (steel on steel).
- Safe, reliable, all-weather and energy efficient; less air pollution per tonne-km.
- Can carry large passenger volumes at high speed on one corridor.

Limitations

- High initial cost; fixed routes, so no door-to-door service; needs road feeder links.

2. Permanent Way (Track)

Permanent way is the finished track structure – rails, sleepers, ballast and fastenings – laid on the formation to carry the train load safely. It is called 'permanent' to distinguish it from the temporary track laid during construction.

Components of Permanent Way

- **Rails** – steel girders that guide and carry the wheels.
- **Sleepers** – hold rails at correct gauge and transfer load to ballast.
- **Ballast** – broken-stone bed under sleepers; spreads load, drains water.
- **Fastenings** – fish plates, bolts, spikes, keys, chairs, bearing plates, clips.
- **Formation (sub-grade)** – prepared soil on which ballast rests.

Requirements of an ideal permanent way

- Correct gauge, alignment and level (except cant); uniform and sufficiently **resilient**.
- Adequate strength to bear loads and lateral forces; good drainage; easy renewal.
- Low maintenance cost and provision for expansion and creep control.

Cross-section of permanent way

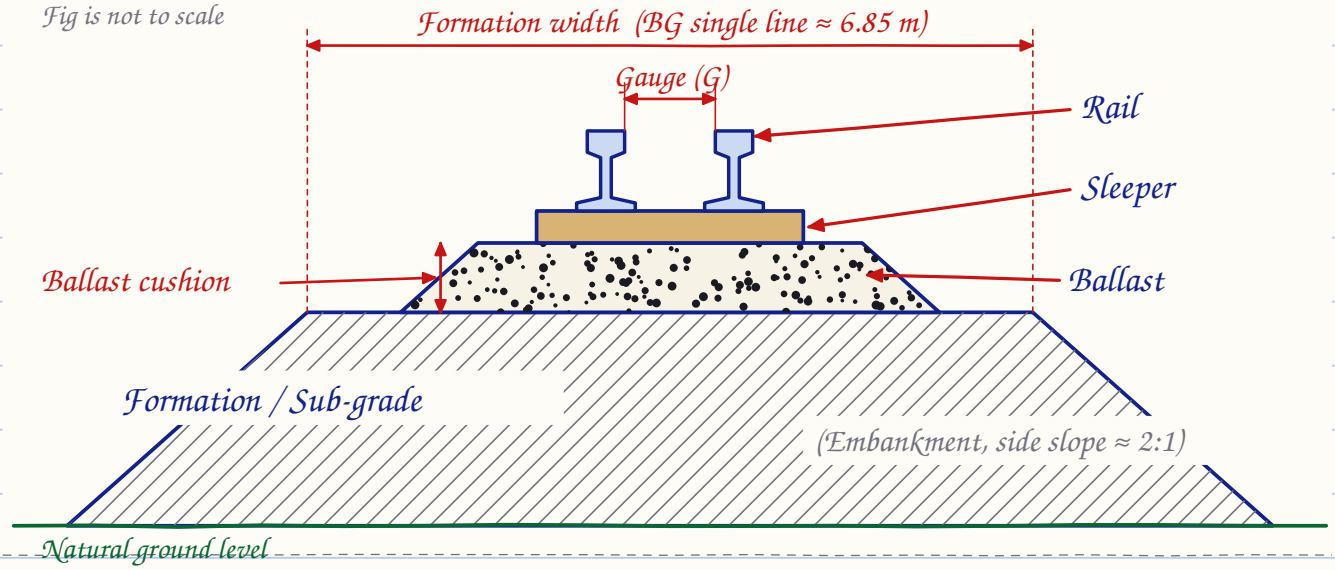


Fig: Cross-section of permanent way (single line, embankment)

- **Formation width** (BG single line embankment ≈ 6.85 m; double line ≈ 11.15 m).
- **Ballast cushion** (below sleeper) 250 – 300 mm; ballast section width ≈ 3.35 m (BG).
- Embankment side slope about 2:1 (H:V); cutting side slope about 1:1 to 1.5:1 – depends on soil.

3. Rails

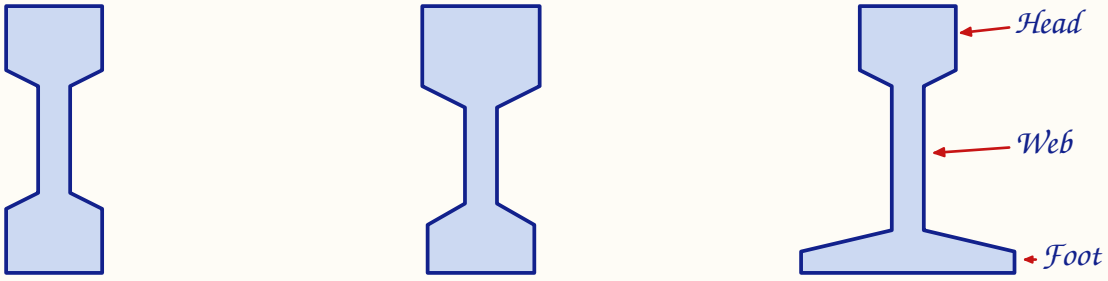
Functions of rails

- Provide a **hard, smooth and continuous surface** for wheels with minimum friction.
- **Guide** the wheels (lateral guidance) and bear the stresses caused by moving loads.
- Transmit the load to the sleepers and ballast over a large area.
- Carry braking and traction forces; act as conductor for track circuits / return current.

Requirements of an ideal rail

- Adequate **vertical stiffness** (depth) and lateral stiffness (head and foot width).
- **Hard head** for wear resistance but tough web/foot against fracture; good fatigue life.
- Weldable, corrosion resistant, low-cost, easy to roll and replace.
- Head wide enough to give wheel contact; foot wide enough for stability.

Rail sections (shape only)



(a) Double headed (b) Bull headed (c) Flat footed

Fig: Types of rail sections

Type	Features	Use
Double headed (DH)	Both heads same, rail can be turned over; needs chairs. Obsolete.	Early railways
Bull headed (BH)	Larger head, smaller foot; cast-iron chairs + keys needed	Britain, sidings
Flat footed (FF) – Vignoles	Wide foot rests directly on sleeper/plate; stiff, cheap, no chairs	Standard in India (~99%)

- Rail sections in India: **90 R (45 kg/m), 52 kg/m, 60 kg/m (UIC 60)** on BG. Heavier rail for higher speed and axle load.
- Standard rail length: **13 m (BG)**, 12 m (MG). Long Welded Rails (LWR) are used for

smooth, low-maintenance track.

- Rail steel: high carbon steel (C = 0.55 – 0.68 %, Mn = 0.7 – 1.0 %); IR uses grade **880** (min UTS 880 N/mm²) and 1080 head-hardened rail.
- Rail defects: wear (vertical/lateral), corrugation, shelling, corrosion at foot, fatigue fracture.

Expansion gap in rails

$$\text{Gap } \Delta = \alpha \cdot L \cdot \Delta T$$

$\alpha = 11.5 \times 10^{-6}$ per °C for steel, L = rail length, ΔT = temperature variation. **Ex:** $L = 13$ m, $\Delta T = 50°C \rightarrow \Delta = 11.5\times10^{-6} \times 13000 \times 50 =$ **7.5 mm**.

4. Sleepers

Functions of sleepers

- Hold the rails to **correct gauge** and keep them in position (alignment & tilt 1 in 20).
- Receive load from rails and **distribute** it over a wider ballast area.
- Act as an elastic medium between rail and ballast; absorb shock and vibration.
- Give lateral stability to the track and keep the rails at proper level (cant on curves).

Requirements of good sleepers

- Initial and maintenance cost low; sufficient bearing area, weight and strength.
- Should allow easy fixing / removal of fastenings and rails; easy to pack with ballast.
- Must resist rot, wear, insects and corrosion; electrically insulating (for track circuits).
- Should be able to be reused; fit for tamping and give good grip with ballast.

Types of sleepers

Type	Merits	Demerits
Wooden (teak, sal, deodar)	Easy handling, resilient, cheap fixing, good insulation, suits all rails	Decay, termite attack, short life (12-20 yr), fire risk
Cast iron (pot, plate, CST-9)	Long life, good scrap value, good gripping	Brittle, heavy, breaks at joints, gauge difficult to hold
Steel (trough, key type)	Light, long life, easy to manufacture, good gauge holding	Corrodes, rail seat loosens, noisy, poor insulation

Type	Merits	Demerits
Concrete (RCC, PSC)	Very long life (40+ yr), heavy, stable, fire-proof, good lateral grip	Heavy, no resilience, needs rubber pad, damages if derailed

- PSC mono-block sleeper (2.75 m, ~270 kg) is the standard on Indian BG main lines; wooden BG sleeper = 2.74 m × 250 mm × 130 mm.

Sleeper density

$$N = M + x$$

N = number of sleepers per rail length, M = rail length in metres, $x = 3$ to 7 (more for heavy/fast traffic). Ex: 13 m rail, $x = 7 \rightarrow N = 20$ sleepers per rail (≈ 1660 per km, spacing ≈ 60 cm).

5. Ballast

Functions of ballast

- Provides a level, hard bed for sleepers; transmits and distributes load to the formation.
- Holds the sleepers in position against lateral and longitudinal movement.
- Gives elasticity and resilience, reducing shock and noise.
- Provides drainage (quick run-off of rain water) and allows packing to adjust track level.
- Checks growth of weeds; helps in maintaining correct super-elevation.

Requirements of good ballast

- Hard, tough, durable, angular with sharp edges (interlocking); no powder/clay.
- Non-porous, should not become slippery when wet; economical and easy to pack (tamping).
- Size ≈ 50 mm (40-65 mm) on main lines; smaller near points & crossings. As per RDSO: Aggregate Impact Value < 20 %, Los Angeles Abrasion < 30 %.

Materials used as ballast

Material	Remarks
Broken stone	Best ballast; used on all main lines (trap, granite, quartzite)
Gravel / river shingle	Rounded, less interlocking; used on branch lines
Cinders / ashes	Cheap and good drainage; corrodes steel, used on sidings

Material	Remarks
Sand	Good drainage, but blows away; used with steel/CI sleepers
Moorum / brick ballast	Poor, low-cost; for lightly loaded tracks, bricks may powder
Selected earth	For temporary track only

- A **sub-ballast / blanket** layer (sand/moorum) is provided on the formation to prevent mud-pumping.

6. Gauges of Track

Gauge = minimum **clear distance between the inner (running) faces** of the two rails of a track.

Standard gauge (Stephenson gauge) = **1435 mm**.

Gauge	Width	Use in India
Broad Gauge (BG)	1676 mm (5'6")	Main lines (≈ 90%+ of network)
Metre Gauge (MG)	1000 mm (3'3⅜")	Branch lines, mostly converted to BG
Narrow Gauge (NG)	762 mm (2'6") / 610 mm (2')	Hill railways, e.g. Darjeeling, Nilgiri, Kalka–Shimla
Standard Gauge	1435 mm	Metro / world standard; used in some Indian metros

Factors in selecting a gauge

- Volume of traffic and speed required; **cost** of construction (wider gauge costs more).
- Physical features – hilly area needs sharper curves, so narrow gauge; future development.
- Uniformity with adjoining systems (Unigauge policy).

Problems of multiple gauges (break of gauge)

- Transhipment of goods and passengers, delay, damage and extra labour cost.
- Extra rolling stock and duplicate workshops/stations needed at junctions; hence the **Unigauge project**.

7. Creep of Rails

Definition

Creep is the gradual longitudinal movement of rails with respect to the sleepers, normally in the direction of dominant traffic. Rails creep forward, not sleepers.

Indications of creep

- Closing of expansion gaps at one end and opening at the other; joints out of square.
- Marks (scratches) on sleepers/ chairs where rail moved; rails shifting past the creep indicator (white paint / stake).
- Kinks in alignment, sleepers out of square, rails shearing fish bolts.

Causes of creep

- Wave action of rail under moving load; drag of wheels; percussion at rail ends.
- Starting, accelerating and braking of trains; temperature changes.
- Loose fastenings, poor ballast, loose joints, sharp curves, steep gradients (more on down gradient).
- Creep is more on heavy-traffic lines, curves, approach to stations, bridges, and with light rail sections.

Theories related to creep

Theory	Explanation
Wave theory	Wheel load makes a depression in the rail like a wave; the wave moves ahead with the wheel and pushes the rail forward, as the rail is elastic between sleepers.
Drag theory	Tractive force from driving wheels (and braking) drags the rail forward by friction between wheel and rail.
Percussion theory	Wheel strikes the end of the next rail (at joint) and gives a forward blow; the energy pushes the rail ahead, so creep is greater at joints.
Temperature (other) causes	Expansion in warm weather and contraction in winter, plus vibration, cause rails to shift.

Effects of creep

- Sleepers get out of square → gauge and alignment disturbed; points and crossings are disturbed.
- Expansion gaps lost → buckling (sun kinks) in summer; joints open up and fish bolts shear.
- Ballast disturbed; increased cost of maintenance; derailment risk.

Prevention / correction of creep

- Pulling back the rails to original position; use anchors / anti-creep devices and good fastenings.
- Keep ballast compact (full), use heavier rail and well-packed sleepers; tighten fish bolts; use long welded rails.
- Use steel / concrete sleepers with elastic clips (Pandrol, ERC) which grip the rail foot.
- Creep is measured with creep indicators; action is needed if creep exceeds about 150 mm (BG).

8. Coning of Wheels and Tilting of Rails

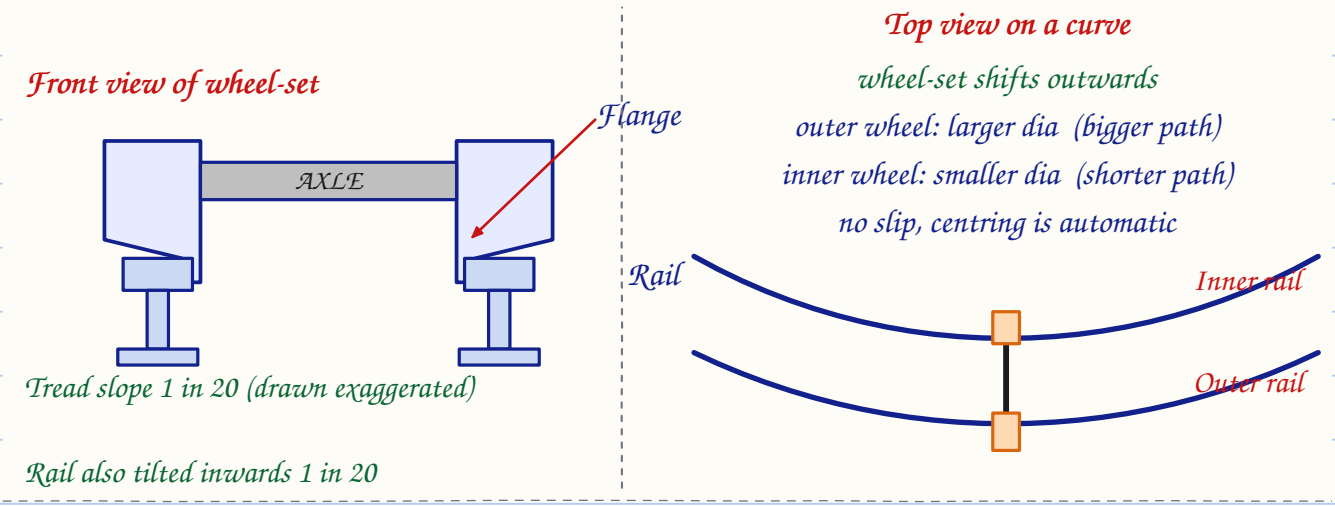


Fig: Coning of wheels and rail tilt

Coning of wheels

Treads of wheels are made conical with a slope of 1 in 20 (not flat) – larger diameter at the flange side. The rails are also tilted inwards 1 in 20 so that the rail head fits the wheel tread.

Reasons / advantages of coning

- Wheel treads wear **uniformly**, which prevents formation of a groove in the rail head.
- Provides **lateral movement** – when the wheel-set shifts to one side, the wheel on that side rolls on a larger diameter and the other on a smaller one, so it **recentres automatically** (self-correcting).
- On curves, the outer wheel gets a larger diameter and travels the longer outer path **without slip**; flange wear and friction are reduced.
- Gives smooth riding, reduces wear of rails and flanges, and checks sway.

Coning on a curve (derivation)

For no slip: $(r + x/20) / (r - x/20) = (R + G/2) / (R - G/2)$, so the lateral shift is:

$$x = 10 \cdot G \cdot r / R$$

Ex: $G = 1.7$ m, wheel radius $r = 0.45$ m, $R = 500$ m $\rightarrow x = 10 \times 1.7 \times 0.45 / 500 = 0.0153$ m $\approx$ **15 mm** (within the flange clearance, so the curve is negotiated without slip).

- If the required shift exceeds flange clearance, flange touches the rail and gauge widening / speed restriction is needed.
- Reason for **tilting rails**: if rails are laid vertical and wheels coned, contact would be at the gauge corner – heavy wear. Tilt 1 in 20 brings contact to the head centre, reduces eccentric load.

9. Adzing of Sleepers

Definition

Adzing is the cutting (levelling) of the rail seat on a wooden sleeper with an **adze** so that the seat slopes **1 in 20 inwards**, giving the required **tilt to the rail**.

- Done only on **wooden sleepers** (the tilt is built in the shape of steel/CI/concrete sleepers and plates).
- Seat cut is just enough to take the rail foot / bearing plate; seats of both rails are cut to slope towards each other.
- Alternative: use **canted bearing plates** (cast / MS plates with 1 in 20 slope) – then adzing is not necessary.

- Disadvantage: adzing removes the hard top layer of the sleeper, so it weakens the sleeper, makes it rot early and shortens its life; hence timber is treated (creosote) after adzing.

10. Rail Fastenings

Fastenings hold the rails to each other (rail joints) and rails to sleepers. They should be strong, simple, elastic and easy to remove.

(a) Fastenings at rail joints

- Fish plates – two per joint, bolted through web to join rail ends; 4-hole (and 6-hole for heavy rail), slope of 1 in 4 matches fish-plate seat.
- Fish bolts (with nuts and spring washers) – 4 per joint; keep the plates tight and allow expansion.
- Joints are suspended, supported or bridge; welding (SWR, LWR/CWR) removes joints.

(b) Fastenings between rail and sleepers

Fastening	Purpose / Note
Dog spike / round spike	Drives into wooden sleeper; holds rail foot or plate (rail-sleeper); screw spike gives better grip
Fang bolt & hook bolt	Used with steel / trough sleepers to fix rail
Keys (wooden, steel)	Wedge to hold rail in the chair (BH rails), with chairs
Chairs (cast iron)	Support the bull-headed rail on sleepers
Bearing plates (MS / CI)	Spread load over a larger sleeper area, give 1 in 20 cant, avoid adzing
Elastic fastening (Pandrol clip, ERC, rubber pad)	Grips the rail foot with spring force, resists creep, resilient; used on PSC sleepers
Anchors / anti-creep	Fix rail to sleepers to resist creep

11. Rail Joints and Welded Rails

A **rail joint** is the place where two rail ends are connected with fish plates. It is the **weakest point** of the track – it causes noise, impact (hammering), wear, creep and needs frequent maintenance.

Type of joint	Description
Supported joint	Rail ends rest directly on one sleeper (joint sleeper); sleeper gets heavy blows.
Suspended joint	Rail ends lie between two sleepers (placed close, 300-450 mm apart); most common on IR; gives elasticity.
Bridge joint	Rail ends joined by a metal bridge plate fixed on two or more sleepers; sturdy but costly.
Staggered / square joint	Staggered: joints of two rails not opposite; square: joints exactly opposite each other (used in India).

Welded rails

- **SWR** (short welded rail, 3 rails = 39 m), **LWR** (long welded rail, > 250 m) and **CWR** (continuous welded rail – a whole section).
- Advantages: no joints → smooth ride, less noise, less wear, **lower maintenance cost**, longer life of rail & rolling stock.
- In LWR the central portion is **fully anchored** and stays constant length; breathing lengths (end portions) move; temperature stresses are taken by the rail itself.
- Welding methods: **flash butt**, **thermit (aluminothermic)** and gas pressure welding; rail temperature is fixed (destressing).
- Care needed: buckling risk in summer; creep control by proper fastenings and full ballast.

12. Worked Examples

Example 1 – Expansion gap

Rail length $L = 13$ m; temperature range $\Delta T = 40°C$; $\alpha = 11.5 \times 10^{-6}$ /°C.

Gap $\Delta = \alpha L \Delta T = 11.5 \times 10^{-6} \times 13000 \times 40 =$ **5.98 mm ≈ 6 mm**.

Example 2 – Number of sleepers

Length of track = 1 km, rail length 13 m, $x = 7 \rightarrow N = 13 + 7 = 20$ per rail.

Rails per km = 1000 / 13 = 76.9 ≈ 77. Sleepers = 77 × 20 = **1540** (about 1540–1660 per km depending on density).

Example 3 – Lateral shift on a curve

$G = 1.75$ m, $r = 0.46$ m, $R = 700$ m. $x = 10 G r / R = 10 \times 1.75 \times 0.46 / 700 = 0.0115$ m = **11.5 mm**.

Example 4 – Ballast volume (approx.)

Ballast width at top 2.25 m, bottom 3.35 m, depth 0.25 m, length 1 km.

Area = (2.25 + 3.35)/2 × 0.25 = 0.70 m²; Volume = 0.70 × 1000 = **700 m^3 per km** (sleeper volume is deducted for exact value).

13. Quick Comparison Tables

Point	Wooden sleeper	Concrete sleeper
Life	12-20 years	40+ years
Weight	Light (~ 80 kg)	Heavy (~ 270 kg)
Elasticity	Good	Poor – needs pad
Fastening	Spikes / screws	Elastic clips (ERC / Pandrol)
Adzing	Required (or canted plate)	Not required (slope built in)
Cost / maintenance	Low initial, high upkeep	High initial, low upkeep

Theory of creep	Main idea	Key word
Wave	Rail moves ahead as wave pushes	Wave / depression
Drag	Tractive force of wheel drags rail	Friction

Theory of creep	Main idea	Key word
Percussion	Wheel hits rail end at joint	Impact

Remember (one-line facts)

- Permanent way = rails + sleepers + ballast + fastenings.
- Gauge is measured between inner faces of the rails.
- Creep occurs in the direction of traffic; it is more on down gradients and at braking zones.
- Coning: wheel tread 1 in 20; rail tilt: 1 in 20 by adzing or canted bearing plates.
- Fish plate slope 1 in 4; max permitted gap about 20 mm (13 m rail).

Important Questions – Unit I

Short answer questions (2 marks)

- Define permanent way and list its components.
- What is gauge? Name the gauges used in India.
- What is coning of wheels? What is the slope?
- Define creep. What is adzing?
- Write the sleeper density formula.

Long answer questions (10 marks)

- Draw a neat cross-section of a permanent way and explain the functions of its components.
- Explain the requirements and types of sleepers with merits and demerits.
- What are the functions of ballast? Explain the requirements and materials of ballast.
- Explain creep and the theories related to creep; give causes, effects, and preventive measures.
- Explain coning of wheels and adzing of sleepers; Discuss various rail fastenings.

Quick revision: Gauge BG 1676, MG 1000, NG 762/610, Std 1435 mm | Cone & tilt = 1 in 20 | $N = M + x$ ($x = 3–7$) | Ballast cushion 250–300 mm | Rail length 13 m.

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RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – II : GEOMETRIC DESIGN OF RAILWAY TRACK

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Gradients – Grade Compensation – Cant and Negative Super Elevation – Cant Deficiency – Degree of Curves – Safe Speed on Railway Track – Points and Crossings – Layout and Functioning of Left Hand Turn Out and Right Hand Turn Outs – Station Yards – Signaling and Interlocking

1. Gradients

Gradient is the rate of rise or fall of the track along its length. It is expressed as **1 in n** (1 vertical in n horizontal) or as a percentage: **% = 100 / n**. Ex: 1 in 200 = 0.5 %.

A gradient is provided to (i) follow the natural ground surface, (ii) reduce earthwork, and (iii) provide drainage in yards.

Types of gradients

Gradient	Meaning	Typical value (BG)
Ruling gradient	Maximum gradient on a section which decides the maximum load a locomotive can haul	1 in 150 – 1 in 200 (plains); 1 in 100 (hills)
Momentum gradient	Steeper than ruling; a train can climb it using the momentum gained on the preceding down gradient	Steeper than ruling, short length
Pusher / helper gradient	Steep gradient on hills where an extra (pusher) engine is needed to haul the train	1 in 37.5 (Bhor ghat); 1 in 100 on hills

Gradient	Meaning	Typical value (BG)
Gradient in station yards	Very flat to avoid movement of standing vehicles; need small slope for drainage	Max 1 in 400; min 1 in 1000

- **Ruling gradient** is not necessarily the steepest gradient; it governs the load of the engine for the whole section.
- **Vertical curves** (parabolic) are provided at changes of grade (when algebraic difference exceeds ≈ 0.4 %) with radius about 4000 m on main lines (IR practice).
- A gradient 'rising' or 'falling' in the direction of traffic; 'Pusher' gradients use banking engines.

2. Grade Compensation on Curves

On a curve, the **resistance to the motion of the train increases** (flange friction, wheel slip, lateral pressure). So the gradient on a curve is **reduced (flattened)** to keep the same tractive effort. This reduction is **grade compensation**.

Gauge	Compensation per degree of curve
Broad Gauge (BG)	0.04 % per degree
Metre Gauge (MG)	0.03 % per degree
Narrow Gauge (NG)	0.02 % per degree

$$\text{Compensated gradient} = \text{Ruling gradient (\%)} - 0.04 \times D$$

Example 1

BG, ruling gradient 1 in 150 = 0.667 %, curve of 4°.

Compensation = 0.04 × 4 = 0.16 %.

Compensated gradient = 0.667 − 0.16 = 0.507 % ≈ **1 in 197** (flatter than 1 in 150).

Example 2

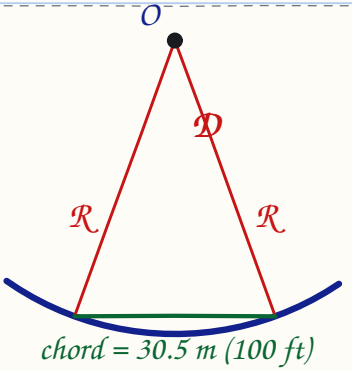
BG, ruling gradient 1 in 100 (1.0 %) and a 5° curve → compensation = 0.2 %.

Gradient on the curve = 1.0 − 0.2 = 0.8 % = **1 in 125**.

- Compensation is applied only where the curve lies on the gradient; **no compensation** is

needed if the gradient is already flatter than the compensated value.

3. Degree of Curve and Radius



Degree of curve (D) :

angle subtended at centre by a chord of 30.5 m length.

$R = 1720 / D$ (metres)

$D = 1720 / R$ (degrees)

Fig: Degree of a curve

D (degree)	1°	2°	3°	4°	5°	6°	8°	10°
R (metres)	1720	860	573	430	344	287	215	172

- Max degree of curve (IR): **BG 10°** ($R \approx 175$ m), MG 16°, NG 40°. In yards, curves are flatter.
- Types of curves: **simple, compound, reverse** curves; **transition curves** connect a straight to a circular curve.

Transition curve

A transition curve (cubic parabola / clothoid) gives a **gradual change** of curvature ($\infty \rightarrow R$) and of cant (0 → full), so the centrifugal force builds up smoothly.

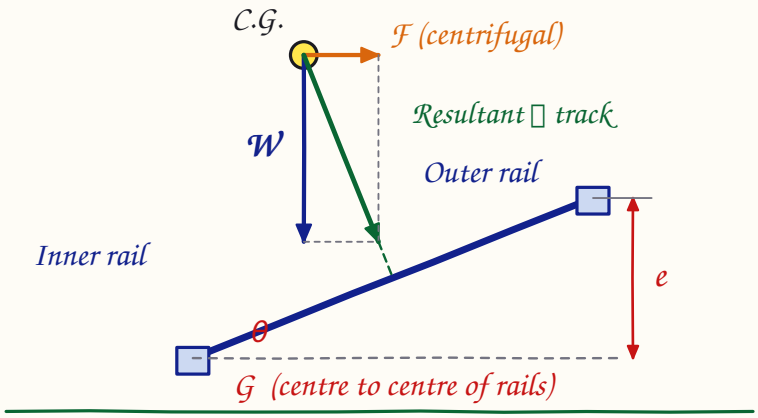
$$L = 0.008\ Ca\ Vm\ ;\ L = 0.008\ Cd\ Vm\ ;\ L = 0.72\ Ca$$

L in m, Ca = actual cant (mm), Cd = cant deficiency (mm), Vm = max speed (km/h). **Take the longest of the three.** Ex: $Ca = 100$, $Vm = 90 \rightarrow L = 0.008 \times 100 \times 90 = 72$ m and $0.72 \times 100 = 72$ m.

4. Cant (Super-elevation)

Definition

Cant (super-elevation) is the raised level of the outer rail over the inner rail on a curve. It is expressed in mm. It balances the centrifugal force $F = mV^2/R$ so that the resultant of W and F is perpendicular to the track.



$\tan\theta = e/G = F/W = V^2/(gR)$

$e = G V^2/(gR)$

In IR units : $e = G V^2/(127 R)$

e (mm), G = 1750 mm (BG), V (km/h), R (m)

$\Rightarrow e \approx 13.78 V^2/R$ mm (BG)

Outer rail raised → balances F

Fig: Cant on a curved track

Need for cant

- To counteract centrifugal force and keep the load shared equally on both rails.
- Prevents overturning of vehicle, side wear of rails and wheel flanges.
- Gives passenger comfort, reduces lateral thrust on track, and gives safe high speed.

Equilibrium cant

$$e = G V^2/(127 R) \quad [mm]$$

G = 1750 mm for BG (centre-to-centre of rails), V in km/h, R in m. So $e = 13.78 V^2/R$ (BG). For MG G ≈ 1058 mm.

- Maximum cant: BG 165 mm (185 mm on some Group A/B routes), MG 90 mm, NG 65 mm; also should not exceed G/10.
- Cant is gradually run out on the transition: cant gradient 1 in 720 (desirable), not steeper than 1 in 360; rate of change ≈ 35 mm/s.
- Equilibrium speed is chosen between goods and fast trains (≈ 3/4 of max speed for mixed traffic).

Example 3

BG curve, $R = 600$ m, equilibrium speed $V = 80$ km/h.

$e = 13.78 \times 80^2 / 600 = 13.78 \times 6400 / 600 =$ **147 mm** (< 165 mm, OK).

Example 4 – speed for a given cant

$R = 600$ m, actual cant given = 100 mm, find equilibrium speed.

$V = 0.27\sqrt{(e\,R)} = 0.27\sqrt{(100 \times 600)} = 0.27 \times 244.9 =$ **66 km/h**.

5. Cant Deficiency and Cant Excess

Cant deficiency (Cd)

The **difference between the equilibrium cant** required for a speed and the **actual cant provided**. Allowed when train speed is higher than equilibrium speed and the rider sits a little 'un-banked'.

- Maximum **cant deficiency**: BG **75 mm** (100 mm on selected high-speed routes), MG 50 mm, NG 40 mm.
- **Cant excess** (Ce) = actual cant − equilibrium cant for the speed of a slow train; max 75 mm; excess causes **inner rail wear**.

$$\text{Equilibrium cant for } V_{max}: \; Ca + Cd = 13.78\, V_{max}^2 / R$$

Negative super-elevation (negative cant)

On a **turn-out of contrary flexure**, the main line is on a curve of cant e, but the branch curves the opposite way. Since both tracks are at the same level, the branch gets the **outer rail lower** than the inner rail – this is **negative cant (Ne)** equal to the cant of the main line there.

- It increases the effective deficiency, so **speed on the branch is restricted**.

$$V = 0.27\sqrt{[\,(Cd(max) - Ne) \times R\,]}$$

Example 5

Branch $R = 300$ m, $Cd(max) = 75$ mm, negative cant $Ne = 20$ mm.

$V = 0.27\sqrt{[(75 - 20) \times 300]} = 0.27 \times 128.4 =$ **34.7 ≈ 34 km/h**.

6. Safe Speed on Railway Track

Safe speed on a curve depends on radius, cant, cant deficiency, length of transition and type of track. The lowest of the speeds below is adopted.

Method	Formula (V km/h)
(i) Cant + deficiency, transitioned curve	$V = 0.27\sqrt{[(Ca + Cd)\,R]}$
(ii) Non-transitioned curve (BG)	$V = 0.27\sqrt{[(Ca + Cd - 75)\,R]}$
(iii) Martin's formula – BG & MG, transitioned	$V = 4.35\sqrt{(R - 67)}$
(iv) Martin's formula – NG / non-transitioned (×0.8 rule)	$V = 3.65\sqrt{(R - 6)}\,(NG)$
(v) Limiting by transition length	from $L = 0.008\,Ca\,V$ (or $Cd\,V$), $V = L/(0.008\,Cd)$

Example 6

BG, $R = 600$ m, $Ca = 100$ mm, $Cd = 75$ mm, transition provided.

(i) $V = 0.27\sqrt{(175 \times 600)} = 0.27 \times 324 = 87.5$ km/h

(iii) $V = 4.35\sqrt{(600 - 67)} = 4.35 \times 23.09 = 100.4$ km/h

Safe speed = lower value ≈ 87 km/h.

- Speeds are also limited by track structure, signals, gradient, vehicle design and sight distance.

7. Points and Crossings

Points (switches) and crossings are arrangements which allow a train to change from one track to another. A turn-out = points + lead rails + crossing.

UNIT – II : Geometric Design of Railway Track

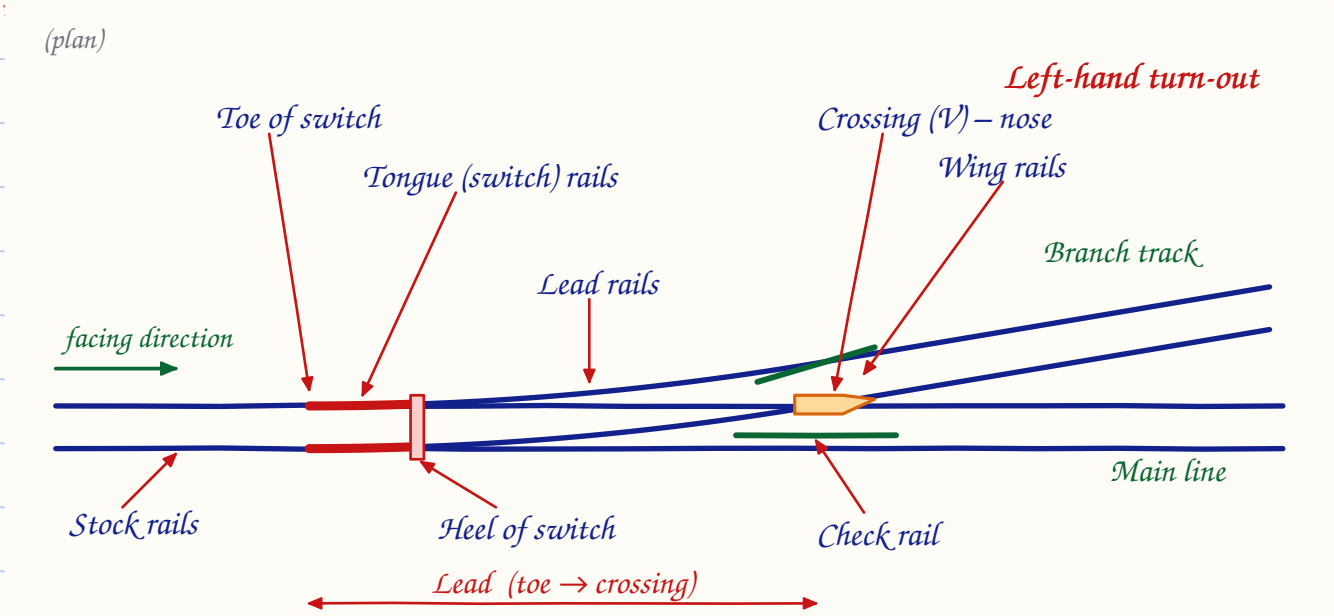


Fig: Layout of a left-hand turn-out

Components

Part	Description
Stock rail	Fixed outer rail against which the tongue rail fits; 2 stock rails (one on each side)
Tongue (switch) rail	Movable tapered rail, pivoted at heel; 2 per switch; moved by lever through stretcher bar; throw ≈ 100-115 mm
Toe / heel of switch	Toe = free thin end of tongue; heel = fixed end (heel block)
Lead rails	Curved rails connecting heel of switch and crossing
Crossing	V-shaped: nose (point rail), wing rails; lets wheel flange cross the other rail; wheel gap gets jumped
Check rails	Placed opposite the nose to guide wheel flange and prevent it from hitting the nose

Number of crossing and angle

Number of a crossing N = 1 in N: the spread is 1 unit for N units of length. Right-angle method: $N = \cot\alpha$; centre-line method: $N = \frac{1}{2}\cot(\alpha/2)$.

Crossing	1 in 8½	1 in 12	1 in 16
Angle α	6° 42′ 35″	4° 45′ 49″	3° 34′ 35″

Crossing	1 in 8½	1 in 12	1 in 16
Use	Yards, slow turnout (≈15 km/h)	Main line turnout (≈30 km/h; up to 50 with curved switch)	Higher-speed turnout

Left hand and right hand turn-outs

- **Left-hand (LH) turn-out** – when a person stands at the **toe** looking towards the crossing (facing direction), the branch diverges to the **left** of the main line.
- **Right-hand (RH) turn-out** – the branch diverges to the **right**. Layout is mirror image; the crossing/ switch parts are the same.
- **Facing point**: train meets toe first; **trailing point**: train enters from the heel / crossing side.

Functioning

- Lever (or point machine) moves the **stretcher bar** which shifts both tongue rails together by the throw.
- For the **main line**: left tongue closes on the left stock rail, right tongue opens; for the **branch**: the other way round.
- Points are **locked** (lock bar / clamp) and **detected** electrically and interlocked with signals so a signal clears only if points are set properly.
- Wheel flange follows the closed tongue, crosses at the crossing guided by check rails, and moves to the diverging track.

Other arrangements

Layout	Description
Crossover	Two turn-outs connecting two parallel tracks (trailing / facing)
Scissors crossover	Two crossovers superimposed, used in confined station yards
Diamond crossing	Two tracks cross each other at an angle (obtuse + acute crossings)
Single / double slip	Diamond + switches inside; allows movement from one track to the other
Triangle	Three turn-outs to reverse a train or engine (wye)

8. Station Yards

A station is a place where trains stop for passenger/goods handling or crossing. A yard is a system of tracks (sidings) laid within defined limits for receiving, sorting, storing and despatching vehicles.

Classification of stations	Description
Halt / flag station	Minor wayside station where only passengers get on/off; trains stop only if needed (flag)
Crossing / Block station	Station where two trains cross or overtake; has loop lines; equipped for block working
Junction	Where two or more lines meet/diverge; Large yards and platforms
Terminal station	End of the line (dead-end or through-terminal); e.g. CSMT, Howrah

Types of yards

- Passenger yard – receives and despatches passenger trains; platforms, coaching depot, washing lines.
- Goods yard – loading/unloading sheds, platforms, cranes, weighbridge.
- Marshalling yard – wagons are sorted, classified and made into new trains. Types: flat yard (shunting by engine), gravity yard, hump yard (wagons run down by gravity after being pushed over the hump).
- Locomotive yard – engine sheds, turn-tables, fuel/water facilities, repair workshops.

Requirements of a good station yard

- On a straight and level (or very flat) track; adequate length and space for expansion.
- Located so as to serve the traffic area, with access roads and proper drainage and signalling.

9. Signalling

Signals are devices that convey instructions to the loco pilot regarding movement of a train, to ensure **safe, quick and orderly** running, to keep trains apart (block system), and to control movement at stations and junctions.

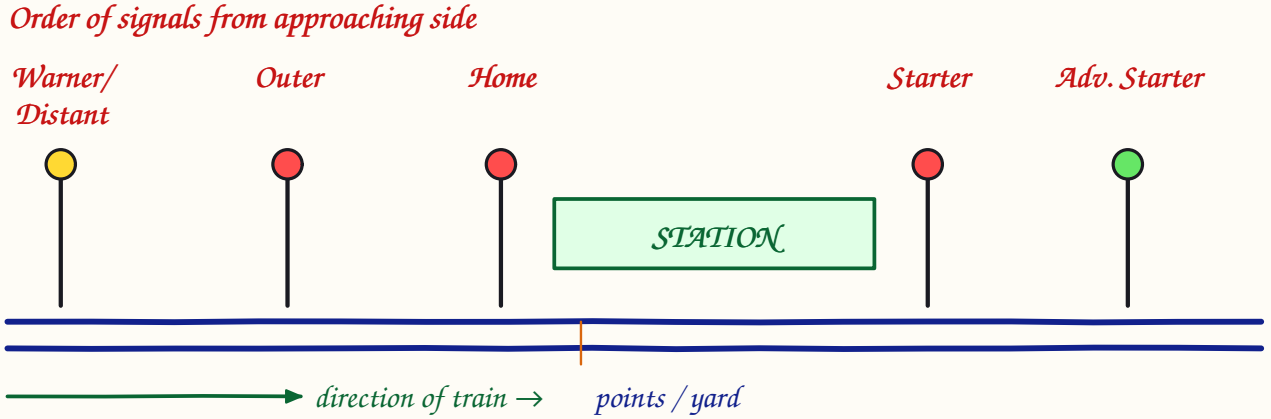


Fig: Fixed signals on a station approach

Classification of signals

Basis	Types
Function	Stop signals (Outer, Home, Starter, Advanced starter, Routing), Warner/Distant signal, Shunting signal
Operation	Mechanical (semaphore arm & wire), Electrical (colour-light, electrically operated)
Mode of display	Semaphore (arm, lower/upper quadrant), Colour-light (2, 3 or 4 aspects)
Method of use	Fixed signals, hand signals (flags/lamps), detonators (audible; for fog & emergency)
Location / Position	Warner/Distant → Outer → Home → Starter → Advanced Starter

Functions of fixed signals

- **Warner / Distant** – warns the driver to be ready to stop at the next stop signal; shows caution (yellow); located about 1 km (braking distance) before.
- **Outer** – first stop signal of a station; protects the station; stands at a safe distance outside Home.
- **Home** – protects station limits / points; driver can enter the platform line only if it is

OFF; stands ≈ 180 m before the first set of points.

- Starter – at the departure end of the platform; authorises train to leave. Advanced starter – last stop signal, at the end of the yard.
- Colour-light aspects: Red = STOP; Yellow = caution (be ready to stop at the next signal); Double yellow = preliminary caution; Green = proceed at full speed.
- Block system: the line is divided into blocks; only one train is allowed in a block at a time (absolute block, automatic block). Modern train protection: Kavach (ATP).

10. Interlocking

Definition

Interlocking is an arrangement of signals, points and other equipment, inter-connected in such a way that their movements are done in a fixed sequence, so that conflicting movements are impossible and a signal can be cleared only when the route is safe.

Objectives / principles

- A signal can be taken 'OFF' only if all points are correctly set and locked for the route and conflicting signals are 'ON'.
- Once signal is OFF, the points cannot be moved (locked) until the signal is put back to 'ON' and the train has passed.
- Two conflicting signals cannot both be OFF; facing points must be locked; the route must be isolated/free (track circuit).

Methods of interlocking

Method	Description
Mechanical (tappet & lock, key)	Levers in a frame with tappets and locking bars; Key interlocking uses transferable keys; Lock-and-block
Electrical / Relay interlocking	Relays and circuits control signals and points; operated from a control panel (route relay interlocking, RRI)
Solid State Interlocking (SSI)	Electronic microprocessor modules replace relays; compact, more reliable, faster

UNIT – II : Geometric Design of Railway Track

Method	Description
Computer-based (EI)	Electronic interlocking with software, centralised traffic control (CTC)

- Interlocking has **Standards I to IV** (higher the standard more the safeguards, used for higher speeds and busier stations).
- Benefits: **safety**, increased line capacity, ease of operation, reduced human errors.

Important Questions – Unit II

Short answer (2 marks)

- What is ruling gradient? Define grade compensation.
- Define cant, cant deficiency and negative super-elevation.
- What is degree of a curve? Write the relation with radius.
- Define turn-out. What is the number of a crossing?
- Write any two functions of signals and what is interlocking?

Long answer (10 marks)

- Explain different types of gradients used on Indian Railways, and grade compensation with an example.
- Derive the formula for equilibrium cant; discuss cant deficiency and negative superelevation.
- Explain the safe speed on curves; solve a numerical on BG.
- With a neat sketch explain the layout of a left-hand turn-out and its functioning.
- Explain station yards, signalling and interlocking.

Quick revision: $e = GV^2/127R = 13.78V^2/R$ (mm) | $V = 0.27\sqrt{[(Ca + Cd)R]}$ | Max Cd BG 75 mm | Grade comp. BG 0.04 %/° | $R = 1720/D$ | $L = 0.008CaVm$.

SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – III : AIRPORT ENGINEERING

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Airport Site Selection – Factors Affecting Site Selection and Surveys – Runway Orientation – Wind Rose Diagram – Basic Runway Length – Correction for Runway Length – Terminal Area – Layout and Functions – Concepts of Terminal Building – Simple Building, Linear Concept, Pier Concept and Satellite Concept – Typical Layouts

1. Introduction to Airport Engineering

An **airport (aerodrome)** is an area of land (or water) with buildings and facilities for the **arrival, departure and ground movement of aircraft**, and for handling passengers and cargo. Air transport is the **fastest** mode for long-distance travel and is vital for trade, tourism and defence.

- **ICAO** – International Civil Aviation Organisation (standards in Annex 14, Aerodromes). **DGCA** – regulator in India; **AAI** – Airports Authority of India builds/operates most airports.
- **Airside**: runways, taxiways, aprons, hangars, control tower, navigational aids. **Landside**: terminal building, roads, car parks, ground transport links.

Classification of airports (ICAO aerodrome reference code)

Code no.	Reference field length	Code letter	Wing span
1	< 800 m	A	< 15 m
2	800 – 1200 m	B	15 – 24 m
3	1200 – 1800 m	C	24 – 36 m
4	≥ 1800 m	D / E / F	36–52 / 52–65 / 65–80 m

- By use: international, domestic, military, STOL / heliport, general aviation; by size and traffic: Category I, II, III etc.

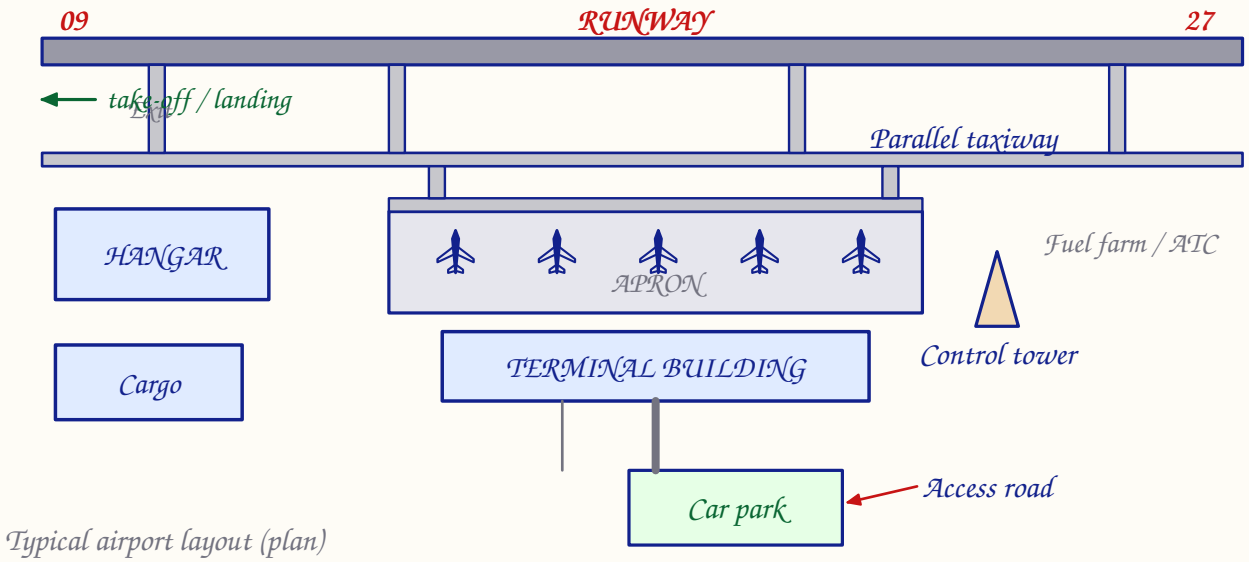


Fig: Typical airport layout

2. Airport Site Selection

Selecting the site is the first and most important step because an airport is costly, permanent and affects a large area. A site is chosen after comparing several alternative sites.

Factors affecting site selection

Factor	Consideration
Regional plan	Airport should fit into the regional / national air-transport plan and the development of the region.
Airport use & traffic	Type of operations (civil, military, cargo), type and number of aircraft, future growth.
Proximity to other airports	Avoid air-space conflict; sufficient separation from other aerodromes.
Ground accessibility	Near the city but not too close; good road/rail links; travel time ≈ 30-45 min.
Topography	Flat, well-drained land; less earthwork; good natural approach funnels; no hills.
Obstructions	Clear approach zones – no tall buildings, towers, chimneys, hills or power lines.

UNIT – III : Airport Engineering

Factor	Consideration
Surrounding land use	Avoid residential areas (noise); Industries, hospitals, schools away from approach.
Noise & environment	Aircraft noise, air pollution, bird activity (landfills, ponds) must be avoided.
Meteorological	Wind direction & speed, visibility (fog, smoke), rainfall, temperature, cloud height.
Soil & drainage	Good bearing soil, high water table avoided; natural drainage; low cost of pavement.
Utilities	Water, electricity, sewage, telecom available at reasonable cost.
Future expansion	Land available for extra runway, terminal & cargo; Economy of land acquisition.

Surveys for airport site

- Topographic / contour survey of the site and surroundings; aerial photographs & Google imagery.
- Soil survey – bearing capacity, CBR, sub-soil water; hydrological survey – rainfall, drainage outfall.
- Meteorological survey – wind data for 5-10 years, visibility, temperature, rainfall → wind rose.
- Traffic & economic survey – passengers, aircraft movements, cargo, catchment area; environmental impact assessment (EIA).
- Obstruction survey – check the obstacle limitation surfaces for height of structures and trees.

Obstacle Limitation Surfaces (OLS) – ICAO (Code 4)

Surface	Slope / size
Approach surface	starts at 60 m from threshold; slope 2 % (1 in 50) for first section
Transitional surface	slope 14.3 % (1 in 7) from strip edge and approach edge up to inner horizontal surface
Inner horizontal surface	45 m above aerodrome elevation, radius about 4000 m

Surface	Slope / size
Conical surface	slope 5 % (1 in 20) outwards from inner horizontal, up to 100 m height

3. Runway Orientation and Wind Rose

Aircraft take off and land against the wind (head wind) because it shortens the run and gives stability. So the runway should be oriented along the direction of the prevailing wind, so that the cross-wind component is minimum.

$$\text{Cross-wind component} = V \cdot \sin\theta$$

V = wind speed, θ = angle between the wind and the runway axis. Ex: $V = 40$ km/h, $\theta = 30°$ → cross wind = $40 \times \sin 30° = 20$ km/h; head wind = $40 \cos 30° = 34.6$ km/h.

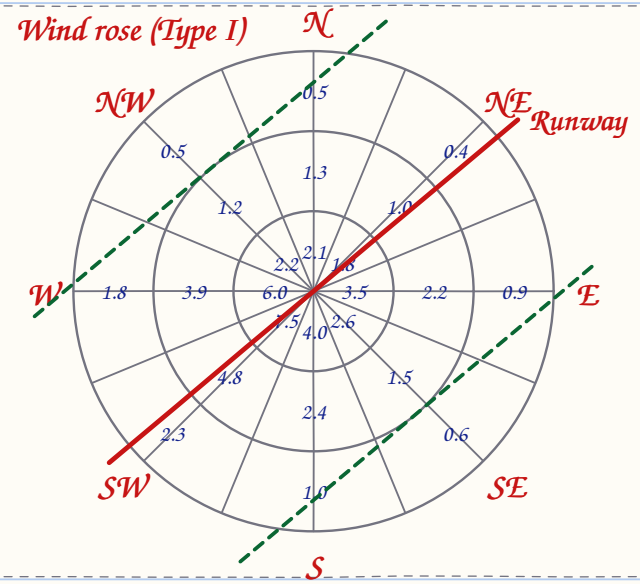
Reference field length	Max permissible cross-wind (ICAO)
≥ 1500 m (Code 4 etc.)	20 knots = 37 km/h
1200 m to 1500 m	13 knots = 24 km/h
< 1200 m	10 knots = 19 km/h

- Wind coverage: percentage of time during which the cross-wind is within the limit. The runway should give at least 95 % coverage; if not, an additional cross-wind runway is needed.

Wind rose diagram

A wind rose is a graphical record of wind direction, intensity and duration at the site. Data are collected for at least 5 – 10 years.

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Use of wind rose

1. Draw 16 radial lines; mark % of time wind blows from each direction (in speed ranges).
2. Take a transparent strip: centre line = runway, side lines at ± permissible cross-wind.
3. Place the strip at the centre of rose and rotate.
4. Add all % lying inside the strip (wind coverage).
5. The position with max total gives best orientation; coverage should be ≥ 95 %.

Red: centre line Green: ± cross-wind limit

ring = wind speed; figures = % of time (illustrative)

Fig: Wind rose with the runway template

Type	Description
Type I	Shows wind direction and intensity (speed). 16 radial lines; numbers on each radial give % of time for different speed ranges. Template of 3 parallel lines is rotated.
Type II	Shows direction, intensity and duration (frequency). Wind speeds in concentric zones of the 8 or 16 directions; used by rotating the template and adding the %.

- Steps (Type I): collect wind data → draw rose → make a transparent strip (centre line + two lines at ± cross-wind limit) → rotate about centre → select the orientation with maximum total % inside the strip.
- Other factors: terrain, obstruction in approach, noise, bird hazard, adjacent airport air space.
- Runway numbering: magnetic azimuth ÷ 10 rounded to the nearest whole; 09 / 27 = 90° and 270°. Parallel runways add L, C, R.

4. Basic Runway Length

Basic runway length is the length required for an aircraft to take off or land under standard conditions, taken from the manufacturer's performance curves, for the critical (most demanding) aircraft.

UNIT – III : Airport Engineering

Standard conditions

- Airport at mean sea level; temperature 15°C (standard atmosphere).
- Runway level (zero gradient); no wind; maximum take-off weight; dry & hard surface.
- No wind, no stopway or clearway; no obstacles; no engine failure.

Corrections to basic runway length (L)

Correction	Rule
1. Elevation	Increase 7 % per 300 m above MSL (air is thinner): $L2 = L\,(1 + 0.07\,h / 300)$
2. Temperature	Increase 1 % for every 1°C that the Airport Reference Temperature (ART) exceeds the standard atmospheric temperature at that elevation: $L3 = L2\,[1 + 0.01\,(ART - Ts)]$
Combined limit	Elevation + temperature correction should not exceed 35 % of L
3. Gradient	ICAO: 20 % for each 1 % effective gradient: $L4 = L3\,(1 + 0.2\,G)$

$$Ts = 15 - 0.0065\,h\,; \quad ART = Ta + (Tm - Ta)/3$$

- h = elevation of the airport in m; Ta = monthly mean of the average daily temperature of the hottest month; Tm = monthly mean of maximum daily temperature of the same month.
- Effective gradient = (max. difference in elevation between highest and lowest point on the runway centre line) ÷ runway length × 100.

Example 1 – Runway length

Basic length = 1800 m, airport elevation = 270 m, ART = 29°C, effective gradient = 0.5 %.

(a) Elevation: $L2 = 1800 \times (1 + 0.07 \times 270/300) = 1800 \times 1.063 =$ **1913.4 m**.

(b) $Ts = 15 - 0.0065 \times 270 = 13.245$°C; $ART - Ts = 15.755$°C → $L3 = 1913.4 \times 1.15755 =$ **2214.9 m**.

Check: 2214.9 / 1800 = 1.23 < 1.35 (OK, limit 35 %).

(c) Gradient: 20 % × 0.5 = 10 % → $L4 = 2214.9 \times 1.10 =$ **2436.4 m ≈ 2440 m**.

Example 2 – Airport Reference Temperature

Ta = 24°C, Tm = 32°C → ART = 24 + (32 – 24)/3 = **26.67°C**.

- Aircraft with higher weight/payload need longer runway; take-off length is usually more than landing length for jet aircraft.

5. Terminal Area

Terminal area is the portion of airport where passengers and cargo are transferred between ground transport and aircraft. It includes the terminal building, apron, car parking, access roads and the support buildings.

Components of the terminal area

- Terminal building – passenger processing: check-in, security, immigration/customs, baggage claim, lounges, shops.
- Apron – place for aircraft to park, load/unload, refuel; with gate positions.
- Hangars (maintenance), control tower, fire station, fuel farm, cargo terminal, and car park / roads.

Functions of terminal area

- Changing mode: ground → air; processing passengers, baggage and cargo; information & communication.
- Providing safe and fast movement of passengers with minimum walking distance; comfort and services.

Aircraft parking configurations on the apron

Type	Remarks
Nose-in	Aircraft parked at 90° to building; nose towards terminal; uses less space, needs tug to push-back
Angled nose-in	Aircraft at an angle; can move out under own power; more apron space
Angled nose-out	Nose pointing away from building at angle
Parallel	Parallel to terminal; needs more frontage; easy movement

Number of gates

$$G = V \cdot T / U$$

G = number of gates, V = design hourly volume (aircraft/hr), T = gate occupancy time (hr), U = utilisation factor (0.6 – 0.8). Ex: $V = 8$, $T = 45$ min $= 0.75$ hr, $U = 0.8 \rightarrow G = 8 \times 0.75 / 0.8 = 7.5 \approx$ 8 gates.

6. Concepts of Terminal Building

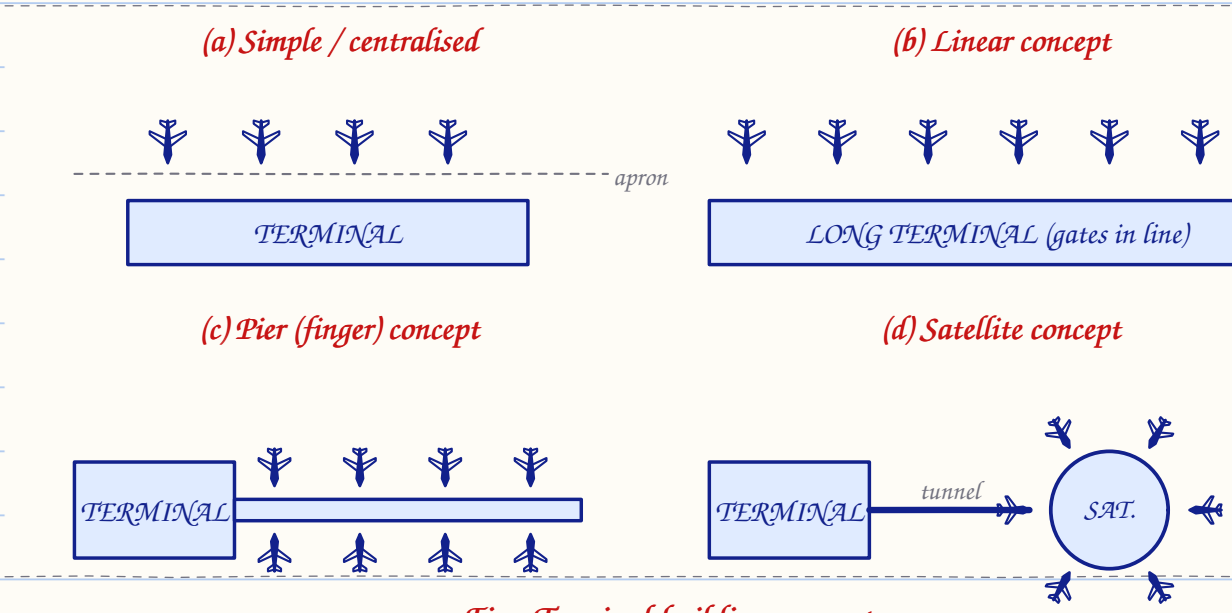


Fig: Terminal building concepts

Concept	Description	Merits	Demerits
Simple (centralised)	Passenger lounge and aircraft close to the single building; aircraft parked on the apron in front	Simple, cheap, short walking, for small airports	Limited expansion; apron crowded
Linear	Building extended linearly; aircraft parked along its length; gates in a line	Short walking, easy expansion in length, direct access	Long distance for transfer if many gates; large frontage
Pier (finger)	A long pier (finger) projects from the terminal; aircraft parked on both sides of pier	More gates on small land; protected passenger walkway	Long walking; aircraft manoeuvre restricted
Satellite	Detached circular/ square satellite building connected to main terminal by underground/ elevated tunnel; aircraft around satellite	Many gates compact; automatic people mover; flexible	Costly; baggage handling complex; long travel for transfer

- Other concept: Transporter (remote apron) – aircraft parked away; passengers carried by buses / mobile lounges. Cheap but less convenient.

7. Typical Layouts of Runways and Airports

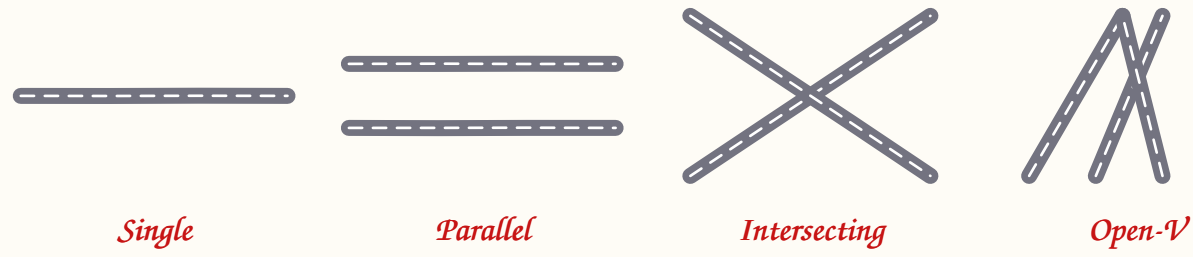


Fig: Runway configurations

- **Single runway** – simplest; capacity limited; used in small airports; **parallel runways** – two runways at a distance; capacity greatly raised (simultaneous landing/take-off).
- **Intersecting runways** – two runways cross; used where winds from different directions; the capacity depends on the crossing point (near the thresholds is better).
- **Open-V runways** – two diverging runways; when wind comes from the closed end, both can be used simultaneously, giving greater capacity.
- Taxiways connect runways with apron; **parallel taxiway** and rapid exit taxiways increase capacity and reduce runway occupancy time.

8. Declared Distances and Runway Terms

Term	Meaning
TORA	Take-Off Run Available – length of runway declared available for ground run of an aircraft taking off
TODA	Take-Off Distance Available = TORA + clearway
ASDA	Accelerate-Stop Distance Available = TORA + stopway
LDA	Landing Distance Available – runway length usable for landing (from displaced threshold)
Stopway	Area beyond take-off run, as wide as runway, able to support an aircraft during an abandoned take-off
Clearway	Obstacle-free area beyond the runway end under control of authority, width ≥ 150 m, slope up to 1.25 %

Term	Meaning
Threshold	Beginning of the usable landing portion of the runway; can be displaced due to obstacles

- Runway length is checked against balanced field length – the distance needed to either continue take-off or stop after an engine failure at the decision speed V1.

9. Wind Coverage – Worked Example

Example 3 – Orientation of a runway

A wind rose (Type I) has been drawn for a site; runway length < 1200 m, so permitted cross-wind = 19 km/h.

Template placed with centre line along N–S (0°): total % of winds inside the strip = 89.0 % (illustrative values).

Rotated to N 20° E: total inside strip = 94.5 %; rotated to N 35° E: 96.2 %; N 50° E: 93.0 %.

Maximum coverage = 96.2 % (> 95 %) → runway orientation: N 35° E (bearing 35° / 215° → number 04 / 22).

Example 4 – Cross-wind check

Wind from 60° to runway at 45 km/h: cross wind = 45 sin 60° = 38.97 km/h (> 37) → not permissible for a Code 4 runway;

the same wind at 40° gives 45 sin 40° = 28.9 km/h (< 37 km/h) → permissible.

- Runways are numbered by magnetic azimuth: a 35° bearing → '04' (rounded 3.5 → 4) at one end and '22' at the other end (215° → 21.5 → 22).

10. Terminal Planning Considerations

Passenger flow in a terminal

Departing: Kerb / car park → check-in (baggage drop) → security → immigration (if international) → departure lounge / shops → boarding gate → aircraft.

Arriving: Aircraft → arrival gate → immigration / health → baggage claim → customs → arrival hall → kerb / transport.

Planning principles

- Short and simple passenger and baggage flow; **separate** departing and arriving flows; clear signs.
- Flexibility to **expand** in modules; segregation of domestic/international and of passengers/cargo.
- Adequate **kerb length**, car parking and a good public transport link.
- Terminal area size is based on **peak-hour passengers** (design busy-hour) and the level of service wanted.

Apron and hangars

- **Apron** pavement is strong (rigid pavement preferred because of fuel spill & heavy load); slope about 0.5 – 1 % for drainage.
- **Hangars**: for storing and maintenance of aircraft; located away from runways and near apron; sized by wing span and tail height; types – arched, cantilever, portal frame.
- **Control tower**: visual control of aerodrome; located to see all runways, taxiways and the apron; Radar (ATC) provides airspace control.

Aircraft stand clearance (concept)

- Wing-tip clearance at a stand: 3 m (code C), 4.5 m (D, E) and 7.5 m (F) as per ICAO, so apron width and gate spacing depend on the **wing span** of the largest aircraft.

11. Comparison and Revision Notes

Point	Simple	Linear	Pier	Satellite
Walking distance	Shortest	Short	Long	Longest
Gates capacity	Low	Moderate	High	Very high
Expansion	Difficult	Easy (length)	Easy (pier)	Easy (new satellite)
Cost	Lowest	Low	Moderate	Highest
Example use	Small airports	Medium airports	Large airports	Major hubs

One-line facts

- Aircraft take-off/landing into the wind; wind coverage ≥ 95 %.
- ICAO cross wind limits: 37 km/h (≥1500 m), 24 km/h (1200–1500 m), 19 km/h (< 1200 m).
- Basic length corrections: elevation → temperature → gradient (in this order).
- ART = Ta + (Tm – Ta)/3; Ts = 15 – 0.0065 h.
- Terminal concept: Simple, Linear, Pier, Satellite (and Transporter).

Important Questions – Unit III

Short answer (2 marks)

- List any four factors affecting airport site selection.
- What is a wind rose? Mention the types.
- State the ICAO limits of cross-wind component.
- What are the standard conditions for basic runway length?
- What is the apron? Name the terminal building concepts.

Long answer (10 marks)

- Discuss the factors that govern the selection of a site for an airport; add notes on surveys.
- Explain the Type I wind rose and how the orientation of runway is found.
- Explain the corrections applied to the basic runway length with a numerical example.
- Explain the layout of the terminal area and its functions.
- Sketch and explain the simple, linear, pier and satellite concepts of terminal building.

UNIT – III : Airport Engineering

Quick revision: Elevation +7 %/300 m | Temp +1 %/°C over standard | Max total 35 % | Gradient +20 % per 1 % | Cross wind 37/24/19 km/h | Coverage ≥ 95 % | $G = VT/U$.

SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – IV : GEOMETRIC DESIGN OF RUNWAYS AND TAXIWA

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Aircraft Characteristics – Influence of Characteristics on Airport Planning and Design – Geometric Design Elements of Runway – Standards and Specifications – Functions of Taxiways – Taxiway Geometric Design – Geometric Elements and Standard Specifications – Runway and Taxiway Lighting

1. Aircraft Characteristics

The design of an airport depends on the characteristics of the aircraft that will use it. The most demanding aircraft is called the critical (design) aircraft.

Characteristic	Meaning / notes
Weight	MTOW (max take-off weight) = OEW (operating empty weight) + payload + fuel. Also MLW (max landing weight), MZFW (max zero-fuel weight)
Size	Wing span, fuselage length, height of tail, wheel base, wheel track (outer main gear wheel span – OMGWS)
Landing gear	Single, dual, dual-tandem, double dual tandem (wide bodies); decides load distribution & tyre pressure
Speed & range	Approach speed, take-off speed, cruise speed; range with full payload
Capacity	Number of passengers / cargo payload (e.g. A320 ≈ 150-180, B777 ≈ 300-400, A380 ≈ 500+)
Turning radius	Minimum radius of turn on ground depends on wheel base and nose-wheel steering angle

UNIT – IV : Geometric Design of Runways and Taxiways

Characteristic	Meaning / notes
Engine type	Piston, turbo-prop, turbo-jet, turbo-fan; decides noise, jet blast and take-off length
Fuel, noise, emissions	Affect environment, land use zones around the airport

Aircraft weight categories (wake turbulence – ICAO)

- Light (L) ≤ 7000 kg; Medium (M) 7000 – 136 000 kg; Heavy (H) > 136 000 kg; and Super (A380). Used for runway separation.

Examples of wing span and ICAO code

Aircraft	Wing span	Code letter
ATR 72 (turboprop)	≈ 27 m	C
A320 / B737	≈ 34 – 36 m	C
B777 / A350	≈ 61 – 65 m	E
A380	≈ 80 m	F

2. Influence of Aircraft Characteristics on Planning and Design

Characteristic	Influence on airport
Weight (MTOW)	Determines pavement thickness, strength (ACN-PCN), runway length, taxiway and apron strength
Wing span	Fixes width of runway, taxiway, apron, separation between taxiways, gate spacing, hangar width
Fuselage length	Decides size of apron gate, hangar length, parking position and turning area
Wheel base & wheel track	Decide turning radius, taxiway width and fillet radius, the pavement edges
Tail height	Decides OLS, height of hangar doors, clearance near approach
Tyre pressure & gear type	Type of pavement (rigid for high pressure), pavement design and ACN

Characteristic	Influence on airport
Speed (landing & take-off)	Runway length, exit taxiway geometry (rapid exit), sight distance, vertical curves
Passenger capacity	Terminal building area, number of gates, parking and road system
Fuel capacity / range	Runway length for long-haul flights, fuel farm and hydrants size
Noise & jet blast	Land-use zoning, blast fences, location of terminal and taxiways

- In short: weight → strength, size → width/clearance, speed → length/curves, capacity → terminal.

3. Geometric Design Elements of Runway

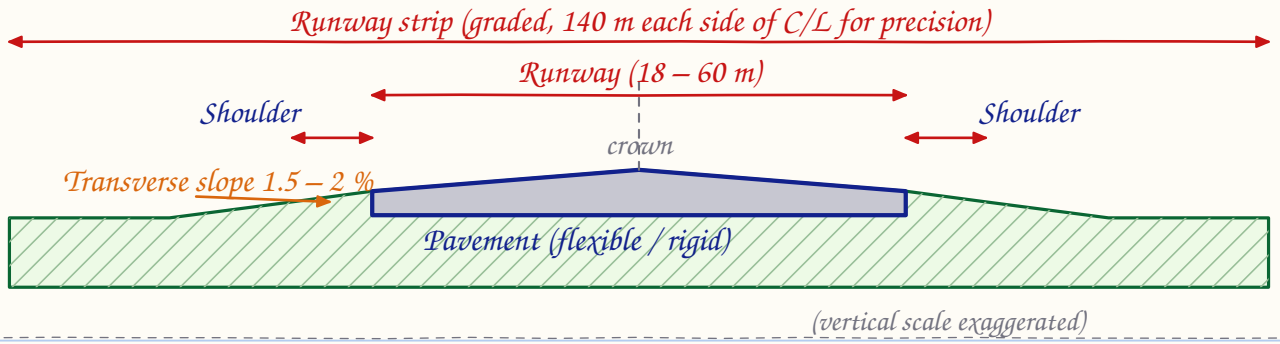


Fig: Cross-section of a runway and strip

(a) Runway length

Obtained from basic length with corrections for elevation, temperature and gradient (Unit III). Length also depends on the critical aircraft, take-off weight and surface (dry/wet).

(b) Runway width

Code number	Width (m)	Remarks
1	18	small aircraft
2	23	
3	30	45 m for code letter D/E
4	45	60 m for code letter F

- Shoulders on both sides, strong enough for occasional run-off and vehicles; overall width

of runway + shoulders ≥ 60 m for code letters D/E and 75 m for F.

(c) Longitudinal gradient

Item	Code 3 and 4	Code 1 and 2
Max effective gradient	1.0 %	2.0 %
Max longitudinal slope	1.25 % (0.8 % in first and last quarter for Code 4)	2 %
Rate of change of slope	0.1 % per 30 m	0.2 % per 30 m
Vertical curve radius (min)	30 000 m (Code 4); 15 000 m (Code 3)	7 500 m
Line-of-sight	3 m above any point must be visible from any other point 3 m above, within half runway length	2 m (Code 2); 1.5 m (Code 1) up to half runway length

- **Effective gradient** = (max. elevation − min. elevation along centre line) ÷ length × 100.

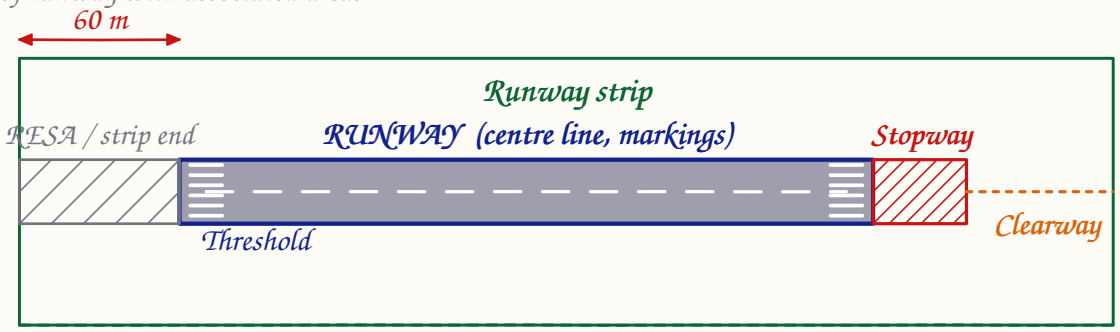
  Example: $L = 2400$ m, difference = 7.2 m → $G = 7.2/2400 \times 100 =$ **0.3 %** (< 1 %, OK).

(d) Transverse gradient

- **1.5 %** for code letters C, D, E, F; **2 %** for A and B (crowned section) to drain water fast; shoulders 2.5 % max. Steeper slopes make aircraft unstable.

(e) Runway strip and safety areas

Plan of runway with associated areas



Strip extends 60 m beyond runway end / stopway; RESA ≥ 90 m (240 m recommended)

Fig: Runway strip, stopway, clearway and RESA

- **Runway strip**: cleared, graded area around the runway to reduce damage to aircraft that run off; width on each side of centre line: **140 m** (precision approach, Code 3/4), **75 m** (non-instrument Code 3/4); extends **60 m** beyond runway end.
- **RESA** (runway end safety area): at least **90 m** (recommended 240 m) beyond strip end,

width ≥ 2 × runway width; graded to stop an aircraft overrunning.

- **Stopway**: same width as runway, strong enough to stop aircraft in an aborted take-off (no landings). **Clearway**: obstacle-free area beyond the runway, width ≥ 150 m, length ≤ half of TORA.

(f) Runway markings

Marking	Description
Designation (number)	Two-digit number = magnetic bearing ÷ 10, e.g. 09, 27 (L/C/R for parallel)
Centre line	Broken white line, 30 m stripe + 20 m gap; gives alignment
Threshold	Series of white longitudinal stripes at the start of landing area
Aiming point	Two broad white bars ~ 300 m from the threshold
Touchdown zone	Pairs of white bars at 150 m interval for precision runways
Edge marking	Continuous white lines on both sides

(g) Pavement

- **Flexible** (bituminous) or **rigid** (concrete, PQC) or composite pavement; designed using **ACN – PCN** method: aircraft is allowed if **ACN ≤ PCN**.

4. Taxiways

Functions of taxiways

- Provide **safe and quick movement** of aircraft between runway, apron, hangar and other areas.
- Allow aircraft to **leave the runway quickly** after landing, so the runway capacity is increased.
- Give access to take-off position, and keep ground movement **separate** from the runway.

Types of taxiways

Type	Description
Parallel (entrance) taxiway	Runs parallel to the runway; used to move aircraft to the runway end; avoids back-tracking

UNIT – IV : Geometric Design of Runways and Taxiways

Type	Description
Exit (turn-off) taxiway	Joins runway at an angle; right-angle exit (90°) or rapid exit / high-speed exit at 30°
Apron taxiway / taxilane	Passes through apron; gives access to aircraft stands
Bypass taxiway	Near the runway end to allow aircraft to pass the waiting ones
Holding bay / holding apron	Area near runway end where aircraft wait for clearance

5. Taxiway Geometric Design

Taxiway turn-offs

V (km/h), f = 0.13; W wheel base, T taxiway width, S = mid main-gear to edge

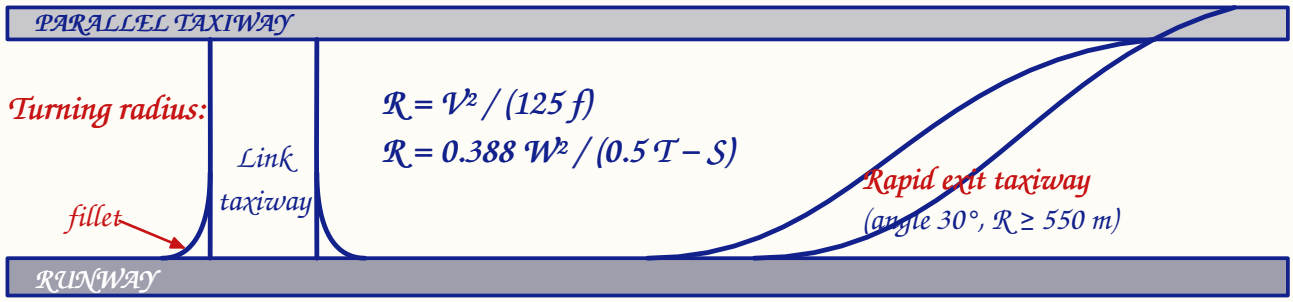


Fig: Taxiway turn-offs and turning radius

Standards (ICAO)

Element	Code letter A, B	Code letter C – F
Longitudinal slope (max)	3 %	1.5 %
Rate of change of slope	1 % per 25 m	1 % per 30 m
Transverse slope	2 %	1.5 %
Sight distance (eye at 1.5 m – 3 m)	visible for 150 m	visible for 300 m (3 m above)
Minimum width of taxiway	A: 7.5 m; B: 10.5 m	C: 15 m (18 m); D: 18 m (23 m); E: 23 m; F: 25 m

- Separation between runway centre line and taxiway centre line (instrument Code 3/4) is about 150 – 180 m, depending on code letter; between taxiways about 1.2 × wing span + clearance.

- **Rapid exit taxiway**: angle **30°**; designed for exit speed **93 km/h** (Code 3/4), turning curve radius ≥ **550 m**, followed by 75 m straight; saves runway occupancy time.
- **Right-angle exit**: 90° with fillets; exit speed ≈ 30 km/h; requires fillet radius.

Radius of turning curve

$$R = V^2 / (125\, f)$$

V = speed of aircraft (km/h), f = coefficient of friction = **0.13** (side friction). Ex: V = 45 km/h → $R = 45^2 / (125 \times 0.13) = 2025 / 16.25 =$ **124.6 ≈ 125 m**.

$$R = 0.388\, W^2 / (0.5\, T - S)$$

W = wheel base (m), T = width of taxiway pavement, S = distance from mid-point of main gear to the edge of pavement. **Ex:** W = 20 m, T = 23 m, S = 5.5 m → $R = 0.388 \times 400 / (11.5 - 5.5) = 155.2 / 6 =$ **25.9 m**. A **fillet** is added at junction to avoid wheels leaving the pavement.

- **Shoulders** on taxiway (stabilised) to prevent erosion by jet blast; overall width of taxiway + shoulders ≈ 25 m (C) to 44 m (E/F).
- Holding positions with **stop bars** are marked before entering the runway.

6. Runway and Taxiway Lighting

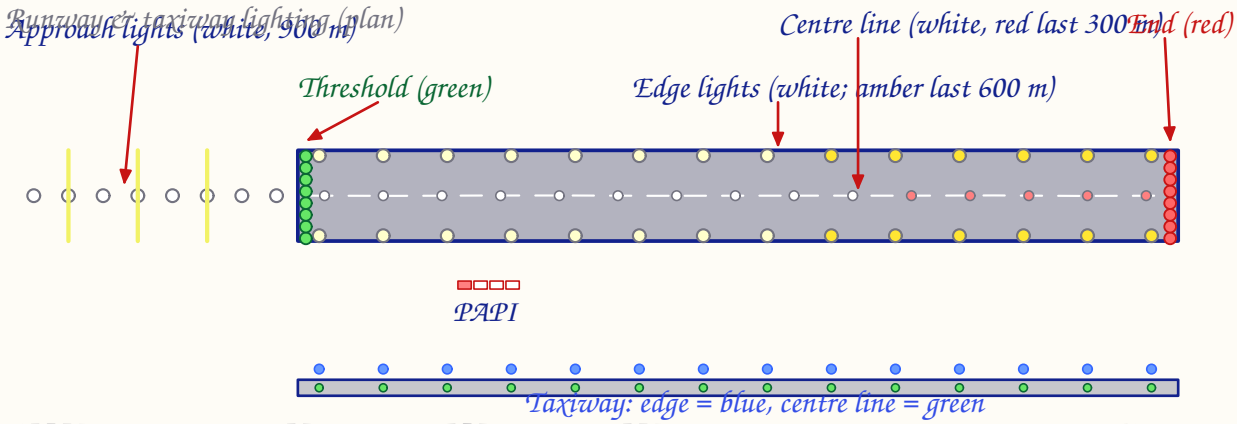


Fig: Airport lighting layout

Light	Colour & arrangement
Approach lighting	Row of white lights along the extended centre line (900 m for CAT I/II/III) with cross-bars (barrettes) – gives direction and distance at night/fog
Threshold lights	**Green** lights across the runway at the threshold (outward-facing)

Light	Colour & arrangement
Runway edge lights	White, along both edges at ≤ 60 m; yellow (amber) for last 600 m / one third (caution zone)
Runway end lights	Red – at the end of runway
Runway centre-line lights	White at 30 m (15 m for CAT III); alternate red-white for next 600 m; red in last 300 m
Touchdown zone lights	White barrettes in the touch-down zone
PAPI / VASI	Visual glide slope: 4 units – white and red lights tell if aircraft is above or below the 3° glide path
Taxiway centre line	Green (for guidance on taxiway); in the exit curves green-yellow for ILS sensitive area
Taxiway edge lights	Blue along edges, at ≤ 60 m
Stop bars / holding	Red across taxiway at holding position; runway guard lights yellow
Aerodrome beacon	Rotating white / green flashing light to identify the airport
Obstruction lights	Red fixed lights on tall objects; apron flood lights for stands

- Lighting requirements: high reliability, intensity control (3 – 5 step), power with secondary source and fast switch-over (e.g. ≤ 1 s for CAT III).
- Taxiway markings are yellow, runway markings are white; instruction signs: mandatory (red/white) and information (yellow/black).

7. Pavement Strength: ACN and PCN

ACN (Aircraft Classification Number) expresses the relative damaging effect of an aircraft on a pavement of a given subgrade strength. PCN (Pavement Classification Number) expresses the bearing strength of the pavement.

- Rule: an aircraft can use the pavement without restriction if ACN ≤ PCN.
- PCN is reported as a code: e.g. PCN 80 / R / B / W / T = number / pavement type (R rigid, F flexible) / subgrade category (A–D) / tyre pressure (W high, X medium, Y low, Z very low) / method (T technical, U using aircraft).

- Subgrade categories: A high, B medium, C low, D ultra-low strength.
- Flexible pavement design for airports: CBR method, FAA method; rigid: Westergaard's stress analysis, PCA method; thickness depends on MTOW, gear type and annual departures.

8. Apron, Holding Bay and Signs

Apron

- Area where aircraft are parked for passenger/cargo loading, fuelling, servicing; adjacent to terminal; strong rigid pavement, slope 0.5 – 1 %.
- Size depends on number of gates, aircraft size (wing span, length) and parking method (nose-in, angled, parallel).
- Wing-tip clearance at stands: 3 m (Code C), 4.5 m (D, E), 7.5 m (F).

Holding bay

- A paved area near the runway end where an aircraft waits for take-off clearance, engine run-up, or to let other aircraft pass; keeps the taxiway free, increases capacity.

Taxiway markings and signs

Item	Description
Taxiway centre-line	Continuous yellow line
Holding position marking	Four lines (two continuous + two dashed) across the taxiway, yellow
Mandatory signs	White on red – runway designation, holding position, no entry
Information signs	Yellow with black letters – taxiway designations, direction, location

9. Worked Examples

Example 1 – Runway pavement width check

Critical aircraft B777 (wing span ≈ 65 m, code 4E). From the table width = 45 m; shoulders 7.5 m each side gives 60 m overall (min. for letter E).

Check: transverse slope = 1.5 % → rise at crown over 22.5 m half-width = 0.015 × 22.5 = 0.34 m.

Example 2 – Rate of change of slope

Two grades meet: +0.6 % and −0.4 % on a Code 4 runway. Change = 1.0 %.

Max rate 0.1 % per 30 m → vertical curve length = 1.0 / 0.1 × 30 = 300 m.

Check with radius: R = 30 000 m per 1 % change → $L = R \times A / 100 = 30\,000 \times 1.0 / 100 =$ 300 m (same).

Example 3 – Taxiway turning radius

Speed on curve 65 km/h, $f = 0.13$: $R = 65^2 / (125 \times 0.13) = 4225 / 16.25 =$ 260 m.

For speed 30 km/h: R = 900 / 16.25 = 55.4 m (right-angle exit).

Example 4 – Longitudinal gradient check

Runway length 2800 m; levels at start = 112.0 m, middle = 114.8 m, end = 113.2 m. Highest − lowest = 114.8 − 112.0 = 2.8 m.

Effective gradient = 2.8 / 2800 × 100 = 0.10 % < 1.0 % → OK.

10. Navigational Aids (Brief)

- ILS (Instrument Landing System): localizer gives lateral guidance, glide path gives vertical guidance (3° slope), marker beacons / DME give distance.
- VOR / DME / NDB radio aids for en-route navigation; GNSS / GAGAN (satellite navigation) for approach with vertical guidance.
- CAT I / II / III precision approaches: decision height ≥ 60 m (CAT I), 30 m (CAT II), lower / zero (CAT III) – approach and centre-line lighting standard increases.
- Windsock / wind direction indicator, signalling area, aerodrome beacon and Runway visual range (RVR) equipment give visual information to the pilot.

11. Summary Tables

Element	Runway	Taxiway
Function	Take-off and landing	Movement between runway, apron, hangars
Longitudinal slope (max)	1 % effective (Code 3/4)	1.5 % (C–F), 3 % (A, B)
Transverse slope	1.5 – 2 %	1.5 – 2 %
Width	18 – 60 m	7.5 – 25 m
Edge light colour	White (amber last 600 m)	Blue
Centre-line light	White / red	Green
Markings	White	Yellow

Remember

- Weight → pavement strength; wing span → widths & separations; wheel base → turning radius.
- Runway edge white, threshold green, end red, taxiway edge blue, taxiway centre green.
- Rapid exit: 30°, exit speed 93 km/h, radius ≥ 550 m.
- ACN ≤ PCN for unrestricted use.

Important Questions – Unit IV

Short answer (2 marks)

- Define critical aircraft, MTOW and wheel base.
- Give two effects of wing span on airport design.
- Write standard transverse slope of runway and ICAO runway widths.
- What is a rapid exit taxiway? Write its angle.
- State the colours of runway edge, threshold, end and taxiway lights.

UNIT – IV : Geometric Design of Runways and Taxiways

Long answer (10 marks)

- Explain aircraft characteristics and their influence on airport planning and design.
- Explain geometric design elements of a runway with ICAO standards.
- Discuss the functions and types of taxiways; explain taxiway geometric design elements.
- Derive the radius of turning curve $R = V^2/(125\,f)$ and solve a numerical.
- Explain the system of runway and taxiway lighting with a neat sketch.

Quick revision: Runway width 18/23/30/45 m | effective gradient ≤ 1 % | transverse 1.5–2 % | RESA 90 m | Strip 60 m beyond | RET 30°, $R \geq 550$ m | $f = 0.13$.

SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – V : PORTS AND HARBOURS

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Harbours – Requirements of Ports and Harbors – Types of Ports – Classification of Harbors – Docks and Types of Docks – Dry Docks, Wharves and Jetties – Breakwaters: Layouts of Different Types of Harbors and Docks – Dredging Operations – Navigation Aids

1. Introduction: Harbours and Ports

A **harbour** is a **sheltered water area** (protected from waves, wind and currents) in which ships can **anchor, load/unload, take shelter and be repaired** safely. A **port** is a harbour **plus** the facilities for transfer of cargo and passengers between sea and land: quays, cranes, warehouses, rail/road links, customs and so on.

- Water transport is the **cheapest** mode for bulk and international trade; about 90 % of world trade (by volume) moves by sea; India has a coastline of about 7500 km.
- **Port = harbour + land-side facilities** (berths, transit sheds, storage yards, container terminals, cranes, tugs, pilotage).
- Ports are managed in India by the **Ministry of Ports, Shipping and Waterways**; 12 **major ports** are under the Central Government and many **minor / intermediate ports** under the State Governments.

Requirements of a good port / harbour

- **Shelter** from waves and winds; calm water inside the harbour.
- **Adequate depth** for the largest ship (draught + under-keel clearance) and sufficiently wide, easily approached **entrance**.
- Good **holding ground** for anchors; enough area (water area for manoeuvring, turning basin

and berthing).

- Good connectivity by rail, road, inland waterways and pipeline to the hinterland.
- Navigation aids (lights, buoys), pilotage, tugs, dredging to keep depth; protection from siltation and littoral drift.
- Facilities: cargo handling equipment, storage, repair docks, fuel and water, medical, customs and security.
- Availability of land for expansion, local construction material, labour; economic viability with traffic potential.

2. Types of Ports

Basis	Types
Location	Sea port (coast), river / inland port, canal port, estuary port
Function / use	Commercial port (general, bulk, container), fishing port, naval (military) port, ferry port, port of refuge, oil / bunker port
Trade	Entrepôt / transhipment port (goods re-exported), ports of call (ships stop for fuel/ supplies), packet station
Ownership (India)	Major ports – under Central Govt. (e.g. Kolkata, Paradip, Visakhapatnam, Chennai, Kamarajar (Ennore), V.O. Chidambaranar, Cochin, New Mangalore, Mormugao, Mumbai, JNPT, Deendayal (Kandla)); minor ports – State

3. Classification of Harbours

UNIT – V : Ports and Harbours

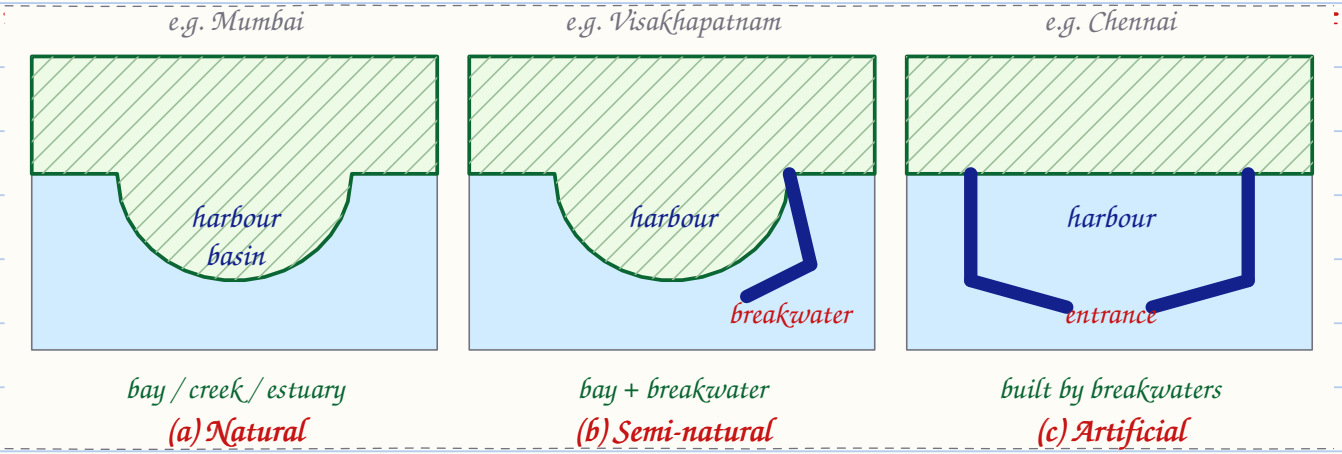


Fig: Natural, semi-natural and artificial harbours

(a) Based on the origin / protection

Class	Description	Remarks
Natural harbour	Formed by nature: bays, creeks, river mouths, estuaries; protected by headlands	Cheapest; limited by nature, e.g. Mumbai
Semi-natural harbour	Protected by nature on sides but opening to sea needs breakwater / jetties or dredging	e.g. Visakhapatnam
Artificial harbour	On an open coast; protection entirely by breakwaters, quay walls and dredged basin	Costly, e.g. Chennai

(b) Based on use / function

- Harbour of refuge – shelter for ships in storms (usually near busy sea routes, e.g. Dover); no trade facilities.
- Commercial harbour – for cargo and passengers; bulk, container, general cargo, oil terminals.
- Fishery harbour – for fishing boats, landing, auction hall, ice plant, repair; shallow depth.
- Naval / military harbour – for warships and submarines; secure, with dry docks and workshops.
- Inland (river) harbour – on rivers/lakes/canals for barges and smaller vessels.

Components of a harbour

- Entrance channel and approach; breakwaters; harbour basin (anchorage) and turning circle; quays, wharves, jetties (berths); docks; navigational aids; transit sheds and warehouses; roads and rail.

4. Tides, Waves and Littoral Drift (Basics)

- **Tides**: rise and fall of sea level by the gravitation of moon and sun. **Spring tides** (new/full moon) have the max range; **neap tides** (quarter moon) the min range. Tidal range decides the need for docks with lock gates.
- **Waves**: wind-generated; wave height H, period T, length $L = 1.56\,T^2$ (deep water, m); wave energy decides the design of breakwaters.
- **Littoral drift**: movement of sand along the shore due to waves hitting at an angle; may cause **siltation of harbour entrance** and erosion elsewhere → needs breakwater layout & dredging.
- **Draught**: depth of ship below water; **harbour depth** = draught + clearance (≈ 1 m or 10 %) + allowance for waves and siltation.

5. Breakwaters

Breakwater is a **barrier** built in the sea to **reduce wave action** and form calm, sheltered water in the harbour; it also protects the entrance from siltation, and may carry a berth or road.

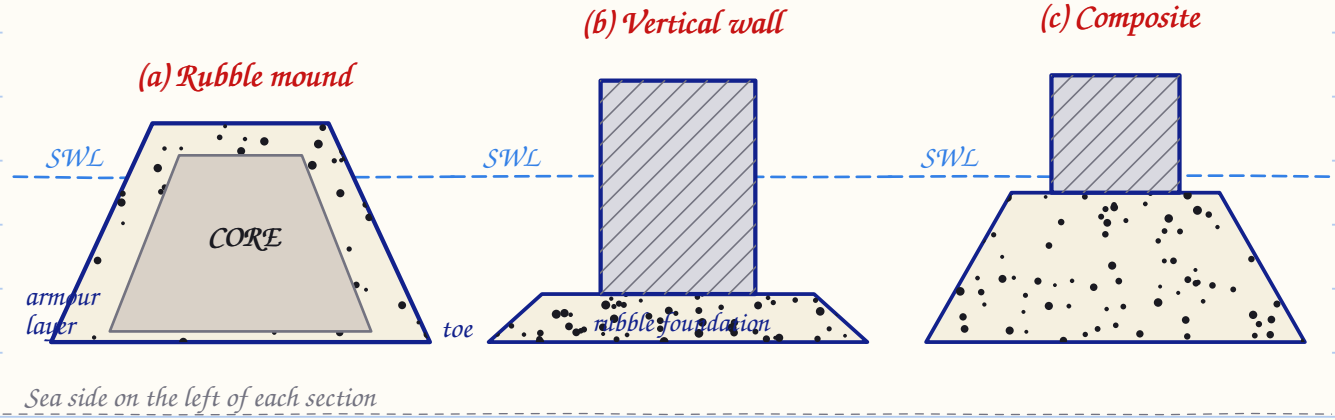


Fig: Types of breakwaters (sections)

UNIT – V : Ports and Harbours

Type	Description	Use
Rubble mound (heap)	Mass of quarried stone dumped with core of small stone and armour layer of heavy stone / concrete blocks (tetrapods, dolos, accropode); waves dissipate energy on the slopes	Most common; shallow to moderate water; flexible foundation
Vertical wall (monolithic)	Concrete blocks / caissons / masonry wall with vertical face on a rubble foundation; reflects waves	Deep water, firm bed; also used as quay
Composite	A low rubble mound topped by a vertical wall; combines both	Medium/deep water; economical
Horizontal composite	Vertical wall with a protective rubble berm / armour on the seaward side	Reduces wave reflection and impact
Special types	Floating (pontoons), pneumatic (air bubbles), hydraulic (water jets) – temporary or small harbours	Small, mild wave area

Layout of breakwaters

- Attached (shore-connected) – one or two arms from shore forming harbour entrance; e.g. Chennai, Visakhapatnam.
- Detached (offshore) – separate to the shore, parallel to coast; protects an open roadstead and reduces erosion.
- Orientation: the entrance is kept away from the direction of heavy waves and littoral drift; arms overlap slightly (staggered) to avoid direct waves entering.

Stability of rubble mound – Hudson formula

$$W = \gamma_r H^3 / [\, K_D (S_r - 1)^3 \cot\theta \,]$$

W = weight of armour unit (kN), γ_r = unit weight of stone (kN/m³), H = design wave height (m), $S_r = \gamma_r/\gamma_w$, θ = angle of slope, K_D = stability coefficient (≈ 4 for rough angular stone, non-breaking, trunk).

Example 1 – Armour stone

$H = 3$ m, $\gamma_r = 26$ kN/m³, $\gamma_w = 10.25$ kN/m³ (sea water), slope 2 H : 1 V ($\cot\theta = 2$), $K_D = 4$.

$S_r = 26 / 10.25 = 2.54$; $(S_r - 1)^3 = 1.54^3 = 3.65$.

$W = 26 \times 3^3 / (4 \times 3.65 \times 2) = 702 / 29.2 =$ 24 kN ≈ 2.4 tonne.

- For vertical wall breakwaters, wave force is found from Sainflou's method (non-breaking, standing wave) or Minikin's method (breaking wave); check sliding, overturning and bearing.

6. Docks and Types of Docks

A dock is an enclosed area of water (or a place) where ships are berthed, loaded/unloaded or repaired, kept at constant level, isolated from tidal changes.

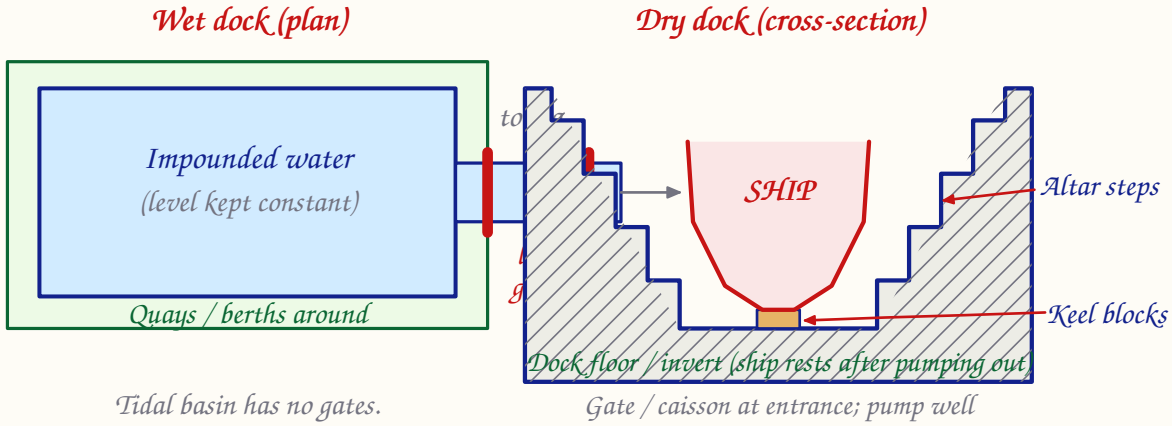


Fig: Wet dock (plan) and dry dock (section)

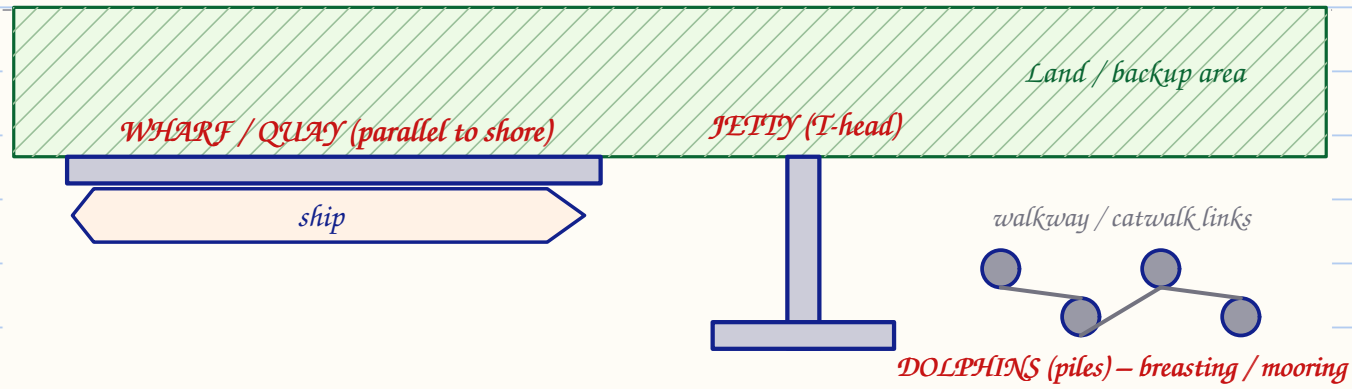
Type	Description
Tidal basin / open dock	Open to the tide; water level varies; no gates; for areas with a small tidal range
Wet dock (impounded dock)	Enclosed basin with lock gates; water kept at high-tide level; ships float at all times; for large tidal range (e.g. Kolkata, London). Entrance through lock with gates and sluices
Dry dock	Used to repair / inspect / paint the hull; can be pumped dry (details below)

Dry docks

- Graving dock (dry dock) – excavated basin with gate (caisson); ship enters at high tide, gate is closed, water is pumped out, and the ship settles on keel blocks; supported by shores; altar steps on sides give access for workers.

- Floating dock – a U-shaped pontoon, partially flooded to sink; ship enters; then water is pumped out and the dock rises with the ship; can be towed to other places.
- Marine railway / slipway (patent slip) – cradle on rails pulls the ship up an inclined slip; for small vessels. Lift dock: ship lifted by hydraulic platform.
- Components of graving dock: entrance, gate or caisson, dock floor (invert), side walls with altars, keel blocks, pumping station (pump well), culverts, capstans, crane and fenders.

7. Wharves, Quays and Jetties



Plan: berthing structures

Fig: Wharf, jetty and dolphins

Structure	Description
Wharf / quay	Structure parallel to the shore along which ships berth; backed by land for cargo handling. Quay = wharf of masonry / concrete wall
Jetty / pier	Structure projecting into the water (perpendicular or inclined to shore) so ships can berth at its sides; open (piles) so water flows; T-head or L-head jetty for tankers
Dolphin	Isolated pile cluster in water: breasting dolphin (takes berthing impact) and mooring dolphin (holds mooring ropes)
Fenders	Rubber/timber cushions on quay face absorbing berthing energy of ship

Types of wharf / quay walls

- Gravity wall (masonry / concrete blocks / caissons); sheet pile wall with anchors; cellular sheet pile; diaphragm wall; open piled deck (RCC deck on piles); relieving platform type.

- Quay face depth = draught + 1 m; width of apron 10-30 m; berth length ≈ ship length + 10 %; wharf crest level above highest tide + 1.5 m.

Berthing energy

$$E = \tfrac{1}{2} \cdot W \cdot V^2 / g \times Cm$$

W = displacement (kN), V = velocity of approach normal to quay (m/s), Cm = added mass coefficient ≈ 1.5 – 1.8. **Ex:** W = 50 000 kN, V = 0.15 m/s, Cm = 1.5 → E = 0.5 × 50000 × 0.0225 / 9.81 × 1.5 = **86 kN·m**.

8. Dredging Operations

Dredging is the process of **excavating** material (silt, sand, clay, rock) from the sea / river bed and transporting it for disposal, to **create** or **maintain** required depth.

- **Purposes**: capital dredging (new channel/basin), **maintenance dredging** (to remove siltation), reclamation (fill land), laying pipes/cables, sand mining, environmental clean-up.

Dredger	Working
Bucket ladder (bucket) dredger	Endless chain of buckets on a ladder digs and lifts material to hoppers; good for hard soils; stationary, anchored
Grab / clamshell dredger	Crane with a grab bucket; suits confined places, quay walls, deep digging
Dipper dredger	A power shovel on a barge; excavates hard soil, rocks and boulders
Suction dredger (plain suction)	Pump and suction pipe lowered to the bed; removes sand and silt as slurry
Cutter suction dredger (CSD)	Rotating cutter head loosens the soil which is sucked; pumps through floating pipeline to reclamation area; for hard soil, channel & land filling
Trailing suction hopper dredger (TSHD)	Self-propelled ship with drag-arm trailing along the bed; loads into **hopper**; sails to disposal area; for maintenance and long channels; no obstruction to traffic

- **Disposal**: dumping at designated sea area, pumping ashore for reclamation (beach nourishment), or use as fill. Environmental checks (turbidity, contaminants) needed.
- Dredging cost depends on soil type, quantity, distance to disposal, and depth; surveys with

echo sounder / bathymetry control depths.

9. Navigation Aids

Navigation aids help the master (captain) to find the position, safe channel and entrance of the port in day, night and fog.

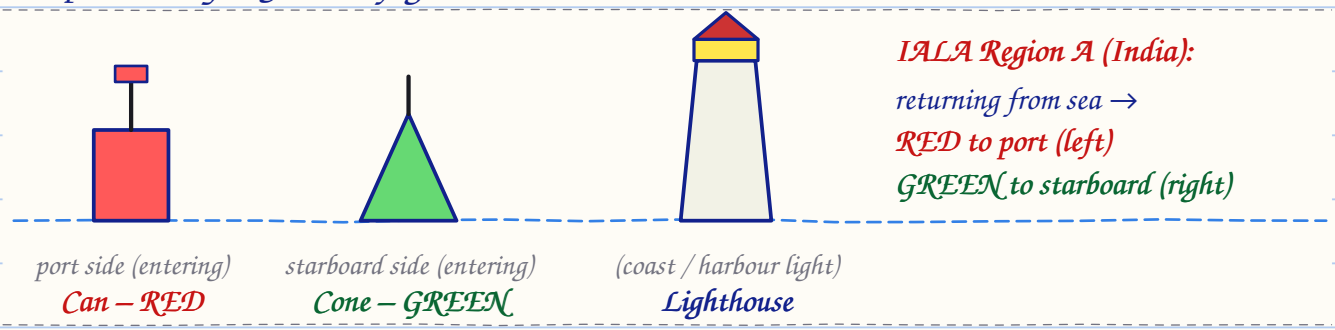


Fig: Navigation marks: buoys and lighthouse

Aid	Function
Lighthouse	Tall tower with a powerful rotating light for coast and harbour identification; visible 20+ nautical miles; each has a characteristic (flash pattern, colour)
Light vessel (lightship)	Anchored ship carrying a light and fog-signal where tower cannot be built (shoals, outer approaches)
Buoys	Floating marks anchored at the bed: lateral (can, cone, spherical, pillar, spar), cardinal, isolated danger, safe water and special marks; colours/shapes show side of channel
Beacons / leading (range) lights	Fixed marks on shore or piles; two lights in line (leading lights) show the centre line of the channel
Fog signals	Sirens, horns, bells, explosive signals, radio beacons used when visibility is poor
Electronic aids	Radar, GPS / DGPS, AIS (Automatic Identification System), VTS (Vessel Traffic Services), ECDIS electronic charts, echo sounder

- IALA Region A buoyage (India, Europe, Africa): entering from the sea, red marks on port (left), green on starboard (right) side.
- Other aids: pilotage, tugs, tide gauges, tide tables, harbour chart, signal stations, and weather warning systems.

10. Layouts of Harbours and Docks

- **Basic harbour layout**: sea → outer approach channel → entrance (between breakwater heads) → outer harbour / turning basin → inner basin with berths → docks / quays → transit sheds → rail & road.
- **Single-arm layout**: one breakwater built from the shore at the up-drift side of littoral drift; cheap, for mild climates.
- **Two-arm (converging) layout**: two arms form a narrow entrance; best protection; entrance oriented away from heavy waves; dredged entrance channel.
- **Dock layouts**: **finger (pier) type** – berths on both sides of parallel piers projecting from the shore; **parallel-quay (basin) type** – quays all round a basin; **marginal quay**; **tidal basin + impounded dock** arranged one behind other with a lock between.
- Width of a turning basin ≈ 2 – 2.5 × length of the largest ship (turning circle); width of entrance ≈ 1.5 – 3 × ship beam to length, so as to restrict waves but allow safe entry (practice varies).

Berth requirements

Item	Typical rule
Berth length	Length of design ship + 10 % (or about 15-30 m clearance for mooring)
Depth at berth	Maximum draught + 1 m under-keel clearance (more in soft bed)
Apron width	Cranes, rails, trucks: 20-40 m for container terminal
Quay crest level	Highest water level + freeboard (≈ 1 – 1.5 m)
Transit shed / yard	Close to the berth, rail and road access

11. Cargo Handling and Coastal Protection

Cargo handling

- **General cargo** by cranes and slings; **bulk cargo** (coal, ore, grain) by grab, conveyors, pneumatic unloaders; **liquid cargo** by pipelines at oil jetties; **containers** by gantry (ship-to-shore) cranes, yard cranes (RTG) and terminal tractors.

- Ro-Ro (roll-on roll-off) ramps for vehicles; passenger terminals for ferries; **Dry port / inland container depot (ICD)** as extension of the sea port.

Coastal protection works (related)

- **Groynes** – built perpendicular to shore to trap littoral drift; **seawalls / revetments** – protect shore from waves; **beach nourishment**; **offshore (detached) breakwaters** reduce wave energy and form tombolos.
- Siltation at harbour entrance is reduced by **sand bypassing** (pump sand from the up-drift side to the down-drift side).

12. Worked Examples

Example 2 – Depth of a harbour basin

Largest ship draught = 12.0 m; under-keel clearance = 1.0 m; allowance for waves and siltation = 0.5 m.

Depth required = 12.0 + 1.0 + 0.5 = **13.5 m** below the lowest water level (chart datum).

Example 3 – Wave length

Wave period T = 8 s. Deep-water wave length $L = 1.56\,T^2 = 1.56 \times 64 =$ **99.8 m ≈ 100 m**.

Example 4 – Berth length

Design ship length = 220 m. Berth length = 220 × 1.10 = **242 m ≈ 245 m**. For two berths in line: 220 + 220 + 3 × 15 m (spacing) = 485 m.

Example 5 – Hudson formula for a larger wave

H = 4 m, same stone and slope: $W = 26 \times 4^3 / (4 \times 3.65 \times 2) = 1664 / 29.2 =$ **57 kN ≈ 5.7 t** → weight increases with H^3 (here $(4/3)^3 = 2.37$ times).

Remember

- Wet dock – **lock gates** keep constant water level; dry dock – ship repaired after **pumping out**.
- Jetty ⊥ shore; wharf ∥ shore; dolphin = isolated piles.
- Capital dredging = new work; **maintenance dredging** = repeated silt removal.
- Breakwater weight of armour $\propto H^3$.

13. Summary and Quick Revision

Topic	Key points
Harbour	Sheltered water area; natural / semi-natural / artificial
Port	Harbour + facilities; sea, river, commercial, fishing, naval, refuge
Breakwater	Rubble mound, vertical wall, composite; Hudson: $W = \gamma r H^3 / [KD (Sr-1)^3 \cot \theta]$
Docks	Tidal basin, wet dock (lock gates), dry dock (graving, floating, slipway)
Wharf / Jetty	Wharf parallel to shore; jetty projects; dolphins for mooring
Dredging	Bucket, grab, dipper, suction, cutter suction, hopper dredgers
Navigation	Lighthouse, light ship, buoys (red port / green starboard), beacons, radar, GPS

Important Questions – Unit V

Short answer (2 marks)

- Differentiate between a port and a harbour.
- Name the types of harbours based on their origin.
- What is a breakwater? Write the Hudson formula.
- Define wet dock and dry dock; what is a jetty and a dolphin?
- What is dredging? Mention two navigation aids.

Long answer (10 marks)

- Explain the requirements of a good harbour and classify harbours with sketches.
- Explain the types of breakwaters with neat sketches and the layouts of harbours.
- What is a dock? Explain the types of docks with a neat sketch of a dry dock.
- Distinguish wharves and jetties; explain the types of quay walls and berthing structures.
- Explain the dredging operations and types of dredgers.
- Describe the navigation aids used in ports and harbours.

Quick revision: Port = harbour + facilities | natural / semi-natural / artificial | rubble mound, vertical wall, composite | tidal basin, wet dock, dry dock | wharf ∥ shore, jetty □

shore | red to port, green to starboard (IALA A).